Assignment -1

1. WHAT ROLES ARE INCLUDED IN A DATA SCIENCE TEAM?
• Data Analyst – Studies past and current data to extract meaningful insights.
• Data Scientist – Builds predictive models using machine learning.
• Data Engineer – Builds systems through which data is moved and stored.
• Business Analyst – Studies data to derive insights useful for business decision-making.
• Process Analyst – Studies workflows and business operations to improve company efficiency.
• ML Engineer – Builds and deploys machine learning models.
• Data Architect – Designs the overall data structure for a company.
• BI (Business Intelligence) Analyst – Builds reports and visualizes data.
• Product Analyst – Studies user reactions to a company's product and suggests improvements.
• Operations Analyst – Helps companies improve logistics and delivery operations.

2. WHAT IS THE DIFFERENCE BETWEEN A DATA ANALYST AND A DATA SCIENTIST?
A Data Analyst studies existing data to understand what has happened, while a Data Scientist builds models that predict what will happen in the future.

3. HOW DOES A DATA ENGINEER SUPPORT A DATA SCIENCE TEAM?
A Data Engineer supports the team by storing and moving data efficiently. They ensure that clean, accessible data is always available for analysts and scientists to work with.

4. WHY ARE ML ENGINEERS IMPORTANT IN MODERN COMPANIES?
ML Engineers are essential because they build and deploy machine learning models at scale. A practical example is recommendation systems such as how Netflix recommends movies tailored to each user.

5. HOW DO DATA TEAMS HELP QUICK-COMMERCE COMPANIES LIKE BLINKIT?
Data teams help quick-commerce companies by identifying common customer behaviour patterns and optimizing delivery routes — enabling them to deliver products within the 10–15 minute window their model promises.

Assignment - 2

1. WHAT IS THE AVERAGE SALARY OF A DATA SCIENTIST IN INDIA?
• Entry-Level – ₹4L – ₹8L per annum
• Mid-Level – ₹8L – ₹18L per annum
• Senior-Level – ₹15L – ₹30L per annum
• Lead / AI Specialist – ₹30L – ₹60L per annum

2. WHY DO SALARIES DIFFER BETWEEN CITIES?
Salaries vary from city to city primarily due to differences in the cost of living. Metro cities like Bengaluru, Mumbai, and Hyderabad tend to offer higher salaries compared to smaller cities, reflecting the higher living expenses in those locations.

3. WHICH SKILLS INCREASE SALARY IN DATA SCIENCE?
The following skills can significantly boost a data scientist's earning potential:
• Machine Learning (ML)
• Cloud Platforms (AWS, GCP, Azure)
• Deep Learning
• Generative AI
• Big Data technologies
• Data Engineering

4. WHY ARE DATA SCIENTISTS IN HIGH DEMAND TODAY?
Data scientists are in high demand because they help companies make better decisions, solve complex business problems, and predict future trends using data — all of which directly impact business growth and competitiveness.

5. HOW DOES EXPERIENCE AFFECT SALARY IN DATA SCIENCE?
Experience and salary are directly proportional. As a data scientist gains more years of hands-on experience and deeper expertise, their market value and compensation increase accordingly.

Assignment - 3

1. WHAT IS DATA SCIENCE?
Data Science is the process of studying data to find meaningful insights. It helps identify hidden patterns, trends, and relationships within data that can be used to make informed decisions.

2. WHAT ARE THE MAIN STEPS IN A DATA SCIENCE PROJECT?
A typical data science project follows these key steps:
1. Data Collection – Gathering raw data from relevant sources.
2. Data Cleaning – Removing errors, duplicates, and inconsistencies to ensure data quality.
3. Data Analysis – Examining the data to discover patterns and draw conclusions.
4. Data Visualization – Presenting findings through charts, graphs, and dashboards for easier understanding.
5. Decision Making – Using the insights derived to make informed business or research decisions.

3. WHICH TOOLS ARE COMMONLY USED IN DATA SCIENCE?
Some of the most widely used tools and libraries in Data Science include:
• Programming Language: Python
• Data Manipulation: NumPy, Pandas
• Data Visualization: Matplotlib, Seaborn, Power BI, Tableau
• Machine Learning: Scikit-learn

4. WHAT IS THE DIFFERENCE BETWEEN DATA ANALYSIS AND DATA SCIENCE?
A Data Analyst focuses on studying current and past data to identify problems or support better decision-making. Data Science, on the other hand, is a broader process that not only analyzes data but also builds predictive models to uncover deeper insights and forecast future outcomes.

5. CAN YOU GIVE A REAL-WORLD EXAMPLE WHERE DATA SCIENCE IS USED?
Data Science is applied across many industries. Here are a few examples:
• Netflix Recommendation System – Recommends movies and shows based on a user's watch history and preferences.
• Weather Prediction – Predicts upcoming weather conditions by analyzing historical weather data.
• Online Shopping – When a user searches for a laptop, the platform recommends accessories like keyboards and mice, or similar laptops from other brands at better prices.

Assignment - 4

1. WHAT IS STRUCTURED DATA?
Structured data is data which can be stored in rows and columns.

2. GIVE EXAMPLES OF UNSTRUCTURED DATA.
Examples of unstructured data include:
• Images
• Videos
• Audio recordings
• Social media posts

3. WHAT MAKES SEMI-STRUCTURED DATA DIFFERENT FROM STRUCTURED DATA?
Unlike structured data, semi-structured data does not follow a strict table format. Instead, it uses key-value pairs to store information, making it more flexible but less rigid than structured data.

4. WHY IS UNSTRUCTURED DATA DIFFICULT TO ANALYSE?
Unstructured data is difficult to analyse because it is messy, does not follow a defined format, and requires advanced technologies to process it — unlike structured data, which is clean and organised.

5. HOW CAN AI ANALYSE CCTV FOOTAGE DATA?
AI analyses CCTV footage by detecting the following:
• Suspicious activities
• Customer movement
• Theft or security

Assignment-5

1. WHAT IS PYTHON?
Python is a programming language known for its simple and easy-to-read syntax.

2. WHY IS PYTHON EASY TO LEARN?
Python code is simple and readable, making it beginner-friendly and easy to understand compared to other programming languages.

3. WHO CREATED PYTHON?
Python was created by Guido van Rossum in 1991.

4. WHERE IS PYTHON USED?
Python is used across a wide range of fields, including:
- Data Science
- Web Development
- AI & Machine Learning
- Automation
- Software Engineering

5. WHAT IS A VARIABLE IN PYTHON?
A variable is used to store data. It acts as a container that holds a value which can be referenced and used throughout a program.

Assignment - 6

1. WHAT IS A VARIABLE IN PYTHON?
A variable is used to store data. It acts as a container that holds a value which can be referenced and used throughout a program.

2. WHY DO WE USE VARIABLES IN PROGRAMMING?
Variables are used for several reasons:
• To store information for later use.
• To use the same value multiple times without repeating it.
• To perform calculations.
• To make programs easier to understand and well-organised.

3. WHAT ARE THE RULES FOR NAMING VARIABLES?
• A variable name must start with a letter.
• It cannot start with a number.
• Spaces are not allowed in variable names.
• Special characters (such as @, #, $) are not allowed.
• Python reserved keywords cannot be used as variable names (e.g. if, else, elif, for, while, def).

4. WHAT TYPES OF DATA CAN VARIABLES STORE?
Variables in Python can store the following data types:
• Integer – Whole numbers (e.g. 10, -5)
• Float – Decimal numbers (e.g. 3.14, -0.5)
• String – Text data (e.g. "Hello", "Riham")
• Boolean – True or False values

5. CAN THE VALUE OF A VARIABLE CHANGE DURING EXECUTION?
Yes, the value of a variable can change during the execution of a program. This is what makes variables dynamic and flexible.

Assignment -7

1. WHAT IS A IF CONDITION IN PYTHON?

If conditions in python program allows the program to take decisions based on the situation.

2. WHAT IS DIFFERENCE BETWEEN `if`, `elif`, AND `else`?

`if` condition check condition and runs the code if its true.

`elif` condition checked only if the previous `if` or `elif` condition is false .

`else` condition checked only if all the above condition s false. As it is run at least it has not condition.

3. CAN WE WRITE `if ` CONDITION WITHOUT `else`?

yes, `if` condition can be used without `else`.

4. Where are if conditions used in real-world programs?

it is used in student in student test analysis like whether student is passed or failed

in atm machine, it will allow you to withdraw cash to a limit.

5. How does Python evaluate conditions?

python evaluates conditions by checking the result is true or false. If it true it runs the code or if it is false it will skip the code and move next part of code.

Assignment - 8

1. WHAT IS GOOGLE COLAB AND HOW IS IT DIFFERENT FROM JUPYTER?

Google Colab is cloud based notebook but whereas Jupyter is a local notebook.

2. HOW DO YOU OPEN AND CREATE A NOTEBOOK IN COLAB?

There are 3 methods to open google colab:

- Method 1: Directly from website:
  - Go to : [https://colab.research.google.com]
  - Click on **New Notebook**

- Method 2: From Google Drives
  - Open google drive
  - Click **New -> More -> Google Colaboratory**
  - Notebook. Will open

- Method 3: From URL
  - Type in browser: `colab.new`
  - Directly open a new notebook

3. HOW CAN YOU ENABLE GPU IN GOOGLE COLAB?

There are few steps to enable GPU;

1. Click **Runtime**
2. Select **Change runtime type**
3. Choose **GPU**
4. Click Save

4. EXPLAIN CODE CELL AND TEXT CELL.

Code cell is used to write code and execute it. Whereas Text cell is used to write normal text and also supports markdown.

5. WHAT ARE ADVANTAGES OF USING GOOGLE COLAB?

It is beginners friendly.
No setup required
Free access to GPU and TPU.
Easy to share notebooks
Supports Python libraries like numpy, pandas, and TensorFlow

Assignment - 9

1. WHAT IS VS CODE?

VS Code is a free software to write code. It is made by Microsoft.

2 WHY IS VS CODE USED?

It is beginner friendly. It works fast, supports many languages, has extensions and good for projects.

3. HOW DO YOU RUN PYTHON CODE IN VS CODE?

- Install python
- Install python code
- Write code:

```
print( "Hello, Riham”)
```

- Click 'Run' or use terminal

4. WHAT IS EXTENSION IN VS CODE?

It is referred as extra features for vs code.
for eg: Prettier (to format the code)

5. WHAT IS TERMINAL IN VSCODE?

Terminal is built-in command line interface which allows to run the code, install packages and executes system commands directly inside editor without switching to other applications.

Assignment - 10

1. WHAT IS A FOR LOOP IN PYTHON?

It is used to execute a part of code multiple times. It helps to avoid write same code again and again.

2. EXPLAIN `range()` WITH EXAMPLE.

It decides how many times the loops will runs. Loops starts from 0 by default.

For eg: range(5) -> 0,1,2,3,4

range(1,5) ->1,2,3,4

range(1,10,2) -> 1,3, 5, 7, 9

3. CAN WE ITERATE OVER STRINGS USING FOR LOOP?

Yes, strings are iterable in python, so `for` loop will take one items at a time.

for example:

```
char = 'hello'
for i in char:
    print(i)
```

o/p:

```
h
e
l
l
o
```

4. DIFFERENCE BETWEEN FOR LOOP AND WHILE LOOP?

`for` loop is used when we know how many times we need run a part of code.

`while` loop is when we need to run the code again and again until a certain condition becomes false or true.

5. WHAT ARE BREAK AND CONTINUE STATEMENT?

`break` statement exist the loop immediately, skipping all the remaining statement in iteration loop

for eg:

```
for I in range (5):
    if I ==3:
        break
    print(I)
```

o/p:

```
0
1
2
```

`continue` statement skips the current iteration and jumps to the next one

for eg:

```
for I in range(5):
    if I ==3:
       continue
    print(I)
```

o/p:

```
0
1
2
4
```

Assignment - 11

1. WHAT IS THE PURPOSE OF `input()` IN PYTHON?

`input()` allows us to take value from the user, while the program run.

for eg:

```
name=input('Enter your name: ')
print('Hello', name)
```

o/p:

```
Enter your name: Riham
Hello Riham
```

2. WHY DO WE USE `int()` WITH `input()`?

Because `input()` stores any given value in string data type by default, so `int()` is used to convert string into int.

for eg:

```
num=input()
print(num+2)
```

o/p:

```
TypeError

num=int(input('Enter the no  =  '))
print(num+2)
```

o/p:

```
Enter the no = 3
5
```

3. WHAT TYPE DOES `input()` RETURN BY DEFAULT?

it return string data type.

for eg:

```
a = input()
print(type(a))
```

o/p:

```
<class 'str'>
```

4. HOW CAN YOU TAKE MULTIPLE INPUTS FROM USER?

```
a=int(input())
b=int(input())
print('sum = ', a+b)
```

o/p

```
3
4
sum=7
```

5. WHAT HAPPENS IF USER ENTERS TEXT BUT `int()` IS USED?

It will show ValueError.

Assignment - 12

1. WHAT IS THE PURPOSE OF INSTALLING PYTHON?

Installing python means setting up python on your computer so that we can write and run python code.

2. WHAT IS AN IDE AND WHY IS IT USEFUL?

IDE stands for Integrate Development Environment. It help in typing, error checking, and running programs.

3. WHAT HAPPENS IF PYTHON IS NOT ADDED TO PATH?

Computer or pc will not recognise python code.

4. WHAT IS THE DIFFERENCE BETWEEN IDLE AND OTHER IDES?

IDLE is a simple, beginner-friendly editor bundled with Python, while full IDEs offer advanced features like autocomplete, debugging, Git integration, and extensions for real-world development.

5. HOW CAN YOU CHECK IF PYTHON IS INSTALLED PROPERLY?

In terminal or Command Prompt type:

python —version

Assignment - 13

1. WHAT ARE FUNCTION ARGUMENTS?

Arguments are values passed to a function. They used to give input to the function.

for eg:

```
def add1(a,b):
        return a+b
res=add1(3,4)
print(res)
o/p:
    7
```

2. WHAT IS DEFAULT ARGUMENT?

A default value is given in function. If user doesn't pass value, default is used.

for eg:

```
def sub(a=2):
        return a-1
res=sub(5)
print(res)
res1=sub()
print(res1)
o/p:
4
1
```

3. WHAT IS KEYWORD ARGUMENT?

When user pass the argument using the keyword, then the order of the keyword doesn't matter.

for eg:

```
def printAB(a,b):
        print( "a =", a)
        print("b =" , b)
printAB(b = 2, a =3)
o/p:
        a=3
        b=2
```

4. CAN WE MIX DEFAULT AND KEYWORD ARGUMENTS?

Yes, we can mix Default and keyword argument but order of the argument matter.

for eg:

```
def stud(name, grade):
        print(name)
        print(mark)
stud('Riham', grade='A')
o/p:
Riham
A
```

5. WHAT HAPPENS IF DEFAULT VALUE IS NOT PROVIDED?

If no default value is provided, the parameter becomes mandatory and must be passed, otherwise Python raises an error.

Assignment - 14

1. WHAT IS A FUNCTION IN PYTHON?

A function is a block of code that performs a task. It helps to reuse code and keep.

2. WHY ARE FUNCTIONS IMPORTANT?

It avoid repeating code and making program is easy to understand by breaking into small parts.

3. WHAT IS DIFFERENCE BETWEEN PRINT AND RETURN?

Print() displays the value in the screen. While return() throws or return the value it doesn't display the value.

4. WHAT ARE PARAMETERS IN FUNCTION?

It is the input given to a functions.

5. CAN A FUNCTION RUN WITHOUT PARAMETERS?

yes it can.

Assignment - 15

1. What is a list in Python?

A list is a collections of items. The items are stored inside a square backets '[ ]'. Items can be int, text, or mixture of data types.

2. How do you add item to a list?

We can add items to the list using append() or insert().

For example:

```
a=[1, 2, 3, 4]     |   a=[1, 3, 4]
a.append(5)        |   a.insert(1, 2)
print(a)           |   print(a)
o/p:               |   o/p:
[1, 2,  3, 4, 5]   |   [1, 2, 3, 4]
```

3. What is difference between append() and insert()?

`append()` is going to add items from last in list. `insert ()` take 2 parameters as input, it add item according to the given index.

For example:

```
a=[1, 2, 3, 4]     |   a=[1, 3, 4]
a.append(5)        |   a.insert(1, 2)
print(a)           |   print(a)
o/p:               |   o/p:
[1, 2,  3, 4, 5]   |   [1, 2, 3, 4]
```

4. How to remove item from list?

We can use `remove ()` to remove items from the list.

for example:

```
name = [ "Ravi", ”Casper”, ”Riham”]
name.remove(”Ravi”)
print(name)
o/p:
[”Casper”, ”Riham”}
```

5. Can a list store different data types?

yes, it can different data types in one list.

for example:

```
collection = [”Riham”, 22, ”Dhiraj”]
print(collection)
```

o/p:

[”Riham”, 22, ”Dhiraj”]

Assignment - 16

1. What is a dictionary in Python?

A dictionary is a data type that stores the data in key - value pairs. The key - value pairs are stored inside the curly brackets `{}`.

For example:

fav_member = { "name”: ”Casper”, "age”: 3, ”type”: ”cat"}

2. How do you access a value in dictionary?

In dictionary values can be access using there keys.

For example:

fav = { "name”: ”Casper”, "age”: 3, ”type”: ”cat”}

print(fav[”name”])

O/p:

Casper

3. How to add new key–value pair?

We can add a new key–value pair by assigning a value to a new key.

For example:

fav = {"name": "Casper", "age": 3}

fav["color"] = "white"

print(fav)

Output:

{'name': 'Casper', 'age': 3, 'color': 'white'}

4. What is difference between keys() and values()?

Key ()returns all the keys in the dictionary, while values() returns all the values.

For example:

fav = {"name": "Casper", "age": 3, "type": "cat"}

print(fav.keys())

print(fav.values())

Output:

dict_keys(['name', 'age', 'type'])

dict_values(['Casper', 3, 'cat'])

5. Can dictionary keys be duplicate?

No, dictionary keys cannot be duplicate. If duplicate keys are given, the last value will overwrite the previous one.

For example:

fav = {"name": "Casper", "name": "Tom"}

print(fav)

Output:

{'name': 'Tom'}

Assignment - 17

1. What is a lambda function?
A lambda function is quick short function. It does not has name and it is written in one line.
For eg:

```
'''
add = lamb a,b : a+b
print(add(5,3))
#Output:
#8
'''
```

2. How is lambda different from normal function?
Lamba is used for small functions like sort(), map(), filter() where as normal function is used for big functions for example define a feature of a website or app.

3. Where are lambda functions used?
It is used inside small functions like sort(), map(), filter().

4. Can lambda have multiple arguments?
Yes, lambda can have multiple arguments.
For example:

```
'''
name_greet = lambda name, greet: print(greet, name)
name_greet(hello, Casper)
#Output:
# hello Casper
'''
```

5. Why lambda functions are called anonymous?
Because it doesn't has name that why it is called anonymous.

Assignment - 18

1. WHAT IS A LIST IN PYTHON?
A list is a data type used to store multiple items in a single variable. It is ordered and changeable.

2. HOW DO YOU ADD ITEM TO A LIST?
You can add items to a list using methods like append() and insert().
For example:

```
fruits = ["apple", "banana"]
fruits.append("mango")
```

3. WHAT IS DIFFERENCE BETWEEN APPEND() AND INSERT()?
append() adds an item at the end of the list, whereas insert() adds an item at a specific position.

4. HOW TO REMOVE ITEM FROM LIST?
Items can be removed using methods like remove(), pop(), or clear().
For example:

```
fruits.remove("apple")
```

5. CAN A LIST STORE DIFFERENT DATA TYPES?
Yes, a list can store different types of data such as integers, strings, and even other lists.

Assignment - 19

—-

1. What is list in Python?

A list is a data type used to store multiple items in single variable. It is ordered, changeable and allows duplicate value.

—-

2. Difference between append and extend?

append() just single item at the end of the list. While on the other hand extend() add multiple items at the end of the list.

For example:

```
num=[1, 2, 3]
num.append(4)
print(num)
#Output:
[1,2,3,4]

num=[1, 2, 3]
num.extend([4, 5])
print(num)
#Output:
[1, 2, 3, 4, 5]
```

—-

3. What is list slicing?

List slicing is method of extracting a portion of the list/

`Syntax:

list[start:end:step]`

For example:

```
num=[1, 2, 3, 4]
print(num[1:3])
#Output:
[2,3]
```

—-

4. Can list store different data types?

Yes, list can store different data types.

For example:

```
mixed = [1, "hello", 3.5, True]
```

—-

5. What is mutable in Python?

Mutable means the object's value can be changed after it is created.

For example:

```
a = [1, 2, 3]
a[0] = 10
print(a)  # [10, 2, 3]
```

Lists are mutable, meaning you can modify, add, or remove elements.

Assignment - 20

1. What is a set in Python?
A set is a collection of unique items. It is written inside curly brackets `{}`.
For example:
num={1, 2, 3, 4}

2. Why are duplicate values not allowed in set?
Set don't allow duplicate value because they are meant to store only unique values, and Python enforce this using hashing.
For example:
Num = {1, 2, 2, 3, 4}
print(Num)
#Output:
{1, 2, 3, 4}

3. What is difference between list and set?
List is written in square brackets `[]`and it allows duplicate values but Set is written in curly brackets `{} `and it allows duplicate value.
For example:
a =[1, 2, 3, 3, 4]
print('a=', a)
b={1, 2, 3, 3, 4}
print('b=', b)
#Output:
a=[1,2,3,3,4]
b={1,2,3,4}

4. How do you add and remove items from set?
To add items in set we can use add().
For example:
s = {'Riham', 'Danial' }
s.add('Casper')
print(s)
#Output:
{'Riham', 'Danial', 'Casper'}
To remove items in set we can use remove().
For example:
s = {'Riham', 'Danial', 'Casper'}
s.remove('Danial')
print(s)
#Output:
{'Riham', 'Casper'}

5. What are union and intersection in sets?
Union means combining both the sets without duplicating.
For example:
a = {1,2,3}
b = {3, 4, 5}
print(a|b)
#Output:
{1, 2, 3, 4, 5}

Intersection means only the common items in both the sets.
For example:
a = {1,2,3}
b = {3, 4, 5}
print(a&b)
#Output:
{3}

Assignment - 21

1. WHAT IS A TUPLE IN PYTHON?

A tuple is a collection of items similar to a list, but it is written inside round brackets (). Tuples are immutable, which means they cannot be changed after creation.
For example:
number = (1, 2, 3, 4, 5)

2. WHAT DOES IMMUTABLE MEAN?

Immutable in Python means that the object cannot be changed after it is created.

3. WHAT IS DIFFERENCE BETWEEN LIST AND TUPLE?

A list is mutable (can be changed) and is written in square brackets [], whereas a tuple is immutable (cannot be changed) and is written in round brackets ().
For example:

```
a = [1, 2, 3]
a[0] = 10
print(a)
#Output:
[10, 2, 3]
b = (1, 2, 3)
b[0] = 10 → This will give an error
```

4. HOW DO YOU CREATE SINGLE ITEM TUPLE?

A single item tuple is created by adding a comma after the item.
For example:

```
t = (5,)
print(t)
#Output:
(5,)
```

5. CAN WE MODIFY TUPLE VALUES?

No, we cannot modify tuple values because tuples are immutable.

Assignment - 22

1.What are built-in modules?
Built - in modules comes already installed with python. It is read-made function.

2.How do you use math module?
math module used for mathematical calculations like square, pi, etc.
For example,
```
import math
print(math.sqrt(16))
print(math.pi)
#Output
5.0
3.141592653589793
```

3.What does random module do?
It is used for generating random numbers.
For example:
```
import random
print(random.randint(1,10))
print(random.choice([1,2,3]))
#Output
9
3
```

4.How to get current date in Python?
We can use datetime module to get current date in python.
For example:
```
  import datetime
  print(datetime.date.today())
```

5.What is sys module used for?
It is used to get system information or system related information.
For example:
```
import sys
print(sys.platform)
#Output
 3.12.7 | packaged by Anaconda, Inc. | (main, Oct  4 2024, 08:22:19) [Clang 14.0.6 ]
```

Assignment - 23

1. What are Python keywords?
Keyword are special words in python which has fixed meaning. It can't be used as variable.
Example:
if, else, for, while, True, False, etc.

2. What is an identifier?
Identifiers are names given to variables, functions, classes, etc. It helps us identify different items in a program.
For example:
num = [1, 2, 3, 4, 5]
num is a identifier in this case.

3. What are the rules for naming identifiers?
- It contains letters, numbers, and underscore(_).
- Cannot start with a number
- Cannot use keywords.
- Case - sensitive (Age and age are different)

Its rule is very similar to rule of defining variables
For example:
student_name = "Aman"
marks1 = 85
_total = 100

4. Why can't we use keywords as identifiers?
Keyword has fixed meaning so if we used it as identifiers it would give a error.

5. What happens if two identifiers differ only in case?
Since Python is a case-sensitive language, two identifiers that differ only in case are treated as completely different variables.
For example: `Age = 20 age = 25` In this case, Python will create two separate memory locations, one for `Age` and one for `age`.

Assignment - 24

1. WHAT IS THE PURPOSE OF BREAK STATEMENT?

The `break` statement is used to terminate the loop immediately when a certain condition is met. Once the `break` statement is executed, the control of the program flows to the statement immediately following the body of the loop.
For example:
Python

```
for i in range(1, 10):
   if i == 5:
      break
   print(i)
```

#Output: 1 2 3 4

2. HOW IS CONTINUE DIFFERENT FROM BREAK?

While both are control statements, they have opposite effects:

- **Break:** Completely exits the loop.
- **Continue:** Skips the rest of the code inside the current loop iteration and jumps to the next iteration.

For example:
Python

```
for i in range(1, 5):
   if i == 3:
      continue
   print(i)
```

#Output: 1 2 4 (3 is skipped)

3. WHEN DO WE USE PASS?

The `pass` statement is a null operation. It is used as a placeholder when a statement is syntactically required but no action is needed. It is helpful when you want to implement a function or loop later.
For example:
Python

```
def my_function():
   pass

for x in [0, 1, 2]:
   pass
```

4. CAN BREAK BE USED OUTSIDE LOOP?

No, the `break` statement can only be used inside a `for` or `while` loop. If used outside a loop, Python will provide a **SyntaxError**.

5. WHAT HAPPENS AFTER CONTINUE IS EXECUTED?

When `continue` is executed, the program ignores all the statements remaining in the current loop block and immediately returns to the start of the loop to check the condition (for `while`) or move to the next item (for `for`).

Assignment - 25

1. What is list comprehension?
List Comprehension is a short way to create lists. It combines loop and condition in one line.
For example:

```
square = [x*x  for x in range(5)]
print(square)
```

2. How is it different from normal loop?
List Comprehension is a short way to write loop in one line and normal loop will take more than one line.
For example:

List Comprehension	Normal loop
even=[x for x in range(10) if x%2 ==0]	even =[]
print(even)	for i in range(10):
#Output:	if i%2 ==0:
[0, 2, 4, 6, 8]	even.append(i)
	print(even)
	#Output:
	[0, 2, 4, 6, 8]

3. Can we use condition in list comprehension?
Yes, we can use condition in list comprehension.
For example:

```
even=[x  for x in range(10) if x%2 ==0]
print(even)
#Output:
[0, 2, 4, 6, 8]
```

4. Is list comprehension faster than loop?
Yes, it is faster than normal loop.

5. Where is list comprehension useful?
It is useful for creating and performing operations on list.

Assignment - 26

1. What is a module in Python?
A module is a file that contains python code. It can have functions, variables and classes. It helps to reuse the code.

2. Why do we use modules?
Avoid writing same code again and again, it organise program and use ready-made features.

3. What is difference between import and from import?
import just loads the whole library and from import a particular part of the library

4. What is alias in import statement?
In Python, an alias is an alternative name assigned to a module or a specific function during the import process.
For example:
import pandas as pd
import numpy as np
df = pd.DataFrame()

5. Name some built-in Python modules.
Openai, numpy, pandas, matplotlib, math, etc

Assignment - 27

1. What is file handling in Python?
It means working with files in python. It can be used to read from and files and write into files.

2. What is difference between r and w mode?
` r` mode is used to read file and ` w` mode is used to create a file, and add or modify the content in file.
For example:

```
file = open("data.txt", "r")        | file = open("data.txt", "w")
content = file.read()               | file.write("Hello Students")
print(content)                      | file.close()
file.close()                        |
```

3. Why do we use with open()?
We use open() in python to open file.

4. How to read file line by line?
We can read file line by line using ` with ` and ` loop` together.
For example:

```
with open("test.txt", "r") as f:
    for line in f:
        print(line)
```

5. What happens if file is not closed?
The file will remain open if we don't close it. This can lead to memory leaks, data corruption, or errors where other programs cannot access the file.

Assignment - 28

1. What is a class?
A class is a blueprint for creating objects, it defines propertied (data) and methods (functions).

2. What is an object?
An object is an instance of a class. It represents a real - world item.

3. What does the __init__ method do?
The __init__ method is an automatic setup function that gives a new object its starting values (like name or color) the moment it is created.
For example,

```
class Book:
    def __init__(self, title):
        self.title = title

b = Book("Casper")
print(b.title)
#Output
Casper
```

4. What is self in Python?
self is a label that refers to the specific individual object being used, allowing the code to distinguish "my" data from another object's data.

```
class Person:
    def greet(self,name1):
        print("Hello",name1)
p = Person()
p.greet(”Riham")
#Output:
Hello Riham
```

5. Can we create multiple objects from one class?
Yes, you can use one single class (the blueprint) to create an unlimited number of unique objects (the actual items).
For example,

```
class Cat:
    def __init__(self, name):
        self.name = name

c1 = Cat("Casper")
c2 = Cat("Fluffy")
print(c1.name)
print(c2.name)

#Output:
Casper
Fluffy
```

Assignment - 29

1. What is exception handling?
It is used to handle errors in a program. So it stops the whole program from crashing

2. What is purpose of try block?
When we have a doubt that the code can give error, then that code is written inside try. And if the code gives error then except block runs.

3. What does finally do?
It alway get runs regardless the code gives error or not, it gets executed at the end.

4. How do you handle specific errors?
We can handle specific error by using try and except .
For example:

```
try:
   num = 5/0
except ZeroDivisionError:
   print(”A number cannot be divided by Zero")
```

5. What happens if exception is not handled?
It will crash the whole program.

Assignment - 30

1. What is OOP?
OOP stands for Object Oriented Programming is a way of writing programs using objects and classes. It helps to organise code in a better way.

2. What is difference between class and object?
A class is a blueprint for creating objects mean while an object is an instance of class.

3. What is constructor?
It is a special method that runs when the object is created.

4. What are OOP features?
Encapsulation- Group data and methods
Inheritance - Reuse code
Polymorphism - Same method different use
Abstraction - Hide details

5. Why is OOP useful?
It is used to organise code in a better way.

Assignment -31

1. What are NumPy functions?
NumPy functions are built-in tools used to perform fast mathematical and logical operations on arrays. These functions are highly optimized for handling large datasets efficiently.

2. Name some common NumPy functions.
Some commonly used NumPy functions are:
np.array(), np.mean(), np.sum(), np.reshape(), np.zeros(), np.ones(), and np.arange().

3. What is the use of mean()?
The mean() function is used to calculate the average value of the elements in an array. It adds all the elements together and divides by the total count.
For example:

```
import numpy as np
arr = np.array([1, 2, 3, 4, 5])
print(np.mean(arr))
```

4. What is reshape()?
The reshape() function is used to change the shape (dimensions) of an array without changing its data. For example, you can change a 1D array of 6 elements into a 2D array of 2 rows and 3 columns.
For example:

```
arr = np.array([1, 2, 3, 4, 5, 6])
new_arr = arr.reshape(2, 3)
```

5. Difference between zeros() and ones()?
Both functions are used to create new arrays of a specific shape, but they initialize the elements with different values:

- **zeros():** Creates an array where every element is **0**.
- **ones():** Creates an array where every element is **1**.

Assignment -32

1. WHAT ARE PANDAS FUNCTIONS?
Pandas functions are built-in methods used for data manipulation and analysis. They allow us to clean, filter, and transform data stored in structures like Series and DataFrames efficiently.

2. WHAT IS THE USE OF HEAD()?
The `head()` function is used to view the first few rows of a DataFrame. By default, it displays the first 5 rows, which is very helpful for quickly checking the structure and contents of a dataset.
For example:
```
import pandas as pd
df = pd.read_csv("data.csv")
print(df.head())
```

3. HOW TO HANDLE MISSING VALUES IN PANDAS?
Missing values (often shown as NaN) can be handled by either removing them or replacing them with specific values. The most common methods are identifying them with `isnull()`, removing them with `dropna()`, or filling them with `fillna()`.

4. WHAT IS GROUPBY()?
The `groupby()` function is used to split the data into groups based on some criteria. Once grouped, we can apply functions like `sum()`, `mean()`, or `count()` to each group separately to get summarized results.
For example:
```
df.groupby('City')['Sales'].sum()
```

5. DIFFERENCE BETWEEN DROPNA() AND FILLNA()?
Both functions are used to manage missing data, but they take different actions:

- **dropna():** This function removes (drops) any rows or columns that contain missing values.
- **fillna():** This function replaces missing values with a specific value (like 0, the mean, or a string) so that no data is lost.

Assignment -33

1. WHAT IS ND ARRAY?
ND array stands for N-dimensional array. In NumPy, data is stored in arrays of multiple dimensions. These arrays are called ndarray. The "N" means it can have 1D, 2D, 3D, or even more dimensions.

2. WHAT IS NDIM IN NUMPY?
ndim is an attribute in NumPy that is used to find the number of dimensions of an array.

3. DIFFERENCE BETWEEN 1D AND 2D ARRAY?
The primary difference lies in the structure of the data:
- 1D Array: Contains elements in a single line (one row).
- 2D Array: Contains elements arranged in rows and columns (like a table).

4. WHAT IS SHAPE?
The shape attribute is used to know the size of an array. It returns a tuple representing the number of elements in each dimension (for example, a 2D array might have a shape of 3 rows and 4 columns).

5. WHERE ARE ND ARRAYS USED?
ND arrays are used in various fields such as:
- Excel data: Handled as 2D arrays.
- Digital images: Stored as 2D or 3D arrays (pixels and colors).
- Videos: Stored as 4D arrays (sequences of images over time).

Assignment - 34

1. What is a Heat Map?
A heat Map is a type of chart that uses colours to represent data values. It uses different colours show different levels of values. It helps to understand data patterns quickly.

2. Why is Heat Map used in Data Science?
Heat map is used to identify high and low values easily, compare multiple values, understand relationships between data and to make data visualisation simple and attractive.

3. Which library is commonly used for Heat Map?
Heat map is in seaborn library.

4. What is the use of annot=True?
The high value will be shown inside boxes because of annot = True. High value amounts will show darker color. Low value amount will show lighter color.

5. What are the advantages of Heat Map?
It easy to understand, attractive visualisation, fast comparison of data and useful for large datasets.

Assignment - 35

1. What is Matplotlib?
It is a python library used for data visualization. It helps to create charts and graphs easily. It is mainly used in Data Science, Data Analysis and Machine learning.

2. Why is Matplotlib used in Data Science?
It is used to visualise data , easily to compare values, it helps in report and presentations, and making data simple and attractive.

3. What is the difference between bar chart and pie chart?
A bar chart is used to compared value between categories and A pie chart shows percentage distribution.

4. Which function is used to create a line chart?
To create plot line chart we use function plot().

5. What are the advantages of data visualization?
It is easy to learn, support many chart types, useful in data Science projects and world with pandas and numpy

Assignment - 36

1. What is a Scatter Plot?
A sitter plot is a graph used to show the relationship between two variables. It uses small dots to represent data points.

2. Why is Scatter Plot used?
To understand relationships between data, identify trends and patterns, to find outliers in data, and commonly used in data science and ml.

3. Which function is used to create Scatter Plot?
scatter() is used to create scatter plot.

4. What does each dot represent?
Each dot shows one value on the X-axis and one value in Y-axis.

5. What are the advantages of Scatter Plot?
Easy to understand relationships, helps find trends, useful for comparing two variables and simple data visualization

Assignment - 37

1. What is contribution graph in GitHub?
It shows coding activity of the development. It displays how active a developer is on GitHub. The graph used green color boxes to represent daily contributions.

2. Why is contribution graph important?
It shows the consistency of the developer, helps HR to understand developer activity, motivates developers to practice daily and makes GitHub profile look active and professional.

3. What activities count as contributions?
Commits, pull request, issue, repository Creation, and code push.

4. What does dark green color represent?
Dark green color represents high activity.

5. How can developers improve contribution graph?
Developers can improve their contribution graph by:
- Practicing coding daily and pushing their updates to GitHub.
- Breaking large projects down into smaller, daily milestones and committing changes regularly.
- Contributing to open-source projects by fixing bugs or raising issues.
- Documenting their learning journey and maintaining active repositories.

Assignment - 38

1. Why do we create a GitHub account?
Creating a GitHub account means making your personal account on GitHub.
It help you uploads projects, save code, and work with other developers.
A Github account is also useful for building your coding portfolio.

2. What details are required for GitHub signup?
We need internet connections,Mobile no or email, username and password.

3. Why should usernames be professional?
HR checks GitHub profile during interviews.

4. What is the benefit of GitHub for developers?
* Free account available
* Access to open-source projects
* Cloud backup for code
* Better collaboration
* Professional developer identity

5. Can GitHub be used as a portfolio?
Yes it can be used as a portfolio

Assignment - 39

1. What is fork in GitHub?
Fork means creating a copy person's repository into your own GitHub account.
It allows developers to use and modify projects safely without changing the original repository. Fork is mostly used in open-source projects

2. Why do developers use fork?
Fork allows developers to use and modify projects safely without changing the original repository.

3. What is the difference between fork and clone?
Fork create copy on GitHub account, it online process, and mainly used for contribution.
Clone downloads project to local computer, offline/local process and used for development work.

4. How does fork help in open-source contribution?
Forks helps in open-source contributions by adding feature.

5. Can fork affect the original repository
No, it allows safe editing without affecting original code.

Assignment - 40

1. What is GitHub Copilot?
The Github Copilot is an AI coding assistant developed by GitHub and open ai.

2. How does GitHub Copilot help developers?
Copilot helps developers write code faster using AI suggestions.

3. Which editors support GitHub Copilot?
It is supported by many editors, like vs code, jet brains IDEs, Neovim, etc

4. What are the advantages of GitHub Copilot?
Faster coding, Better productivity, Helpful for beginners, reduces manual typing and supports many programming languages.

5. Can GitHub Copilot fully replace developers?
No it cannot replace developer

Assignment - 41

1.What is GitHub profile setup?
GitHub profile setup is the process of customizing your personal page on GitHub. It involves adding a profile picture, a professional bio, linking your social accounts or website, and showcasing your best coding repositories to the public.

2. Why is GitHub profile important for developers?
A GitHub profile serves as a digital resume for developers. It is important because it provides proof of your coding skills, shows your passion for building software, and allows recruiters to directly view the quality of your code and how you collaborate with others.

3. What should a professional GitHub bio contain?
A professional GitHub bio should be clear and concise. It should contain:
Your current role or what you are learning (e.g., "Aspiring AI/ML Engineer").
The core technologies, frameworks, and programming languages you specialize in.
A brief mention of your current projects or goals.
Contact information or links to your LinkedIn profile and personal portfolio.

4. What is contribution graph in GitHub?
The contribution graph is a visual grid on your GitHub profile that tracks your daily coding activity over the past year. It displays your consistency using green blocks—where a block turns green whenever you commit code, open pull requests, or submit issues.

5.How can GitHub profile help in jobs?
A well-optimized GitHub profile helps secure job opportunities by making you stand out to HR managers and technical recruiters. It validates your practical skills beyond what is written on a traditional resume, proves your daily coding consistency, and can attract inbound job offers from tech teams looking for active developers.

Assignment -42

1. What is GitHub?
Github is an online platform used to store, manage, and share code.

2. What is the difference between Git and GitHub?
Git is a local version control software that allows you to track changes in our code and manage project history on your computer.
Github is a cloud-based hosting service that lets you store your git repositories online, making it easy to collaborate and shares code with others.

3. What is a repository in GitHub?
Repository is the project folder stored on Github.

4. Why is GitHub important for developers?
Keeps project backup online, Saves different versions of code, helps developers show their portfolio and companies and hr can check coding projects.

5. What is version control?
Version control is a system that tracks and records changes to files over time, allowing you to recall specific versions later.

Assignment - 43

1. What are pinned repositories in GitHub?
Pinned repositories are the repositories that are shown in the top of the profile.

2. Why should developers use pinned repositories?
To highlight the project, making profile more professional, helps HR quickly checks the skill, Improves portfolio appearance and shows project quality and technology.

3. What type of projects should be pinned?
Web development, AI/ML project, Data Science, Android app, open source contribution

4. How do pinned repositories help in jobs?
Helps HR quickly check skills

5. Why is README important in pinned repositories?
README file makes repositories look professional and helps others to understand the projects.

Assignment - 44

1. What is data?
Data means collections of information or facts.

2. Why is data important in AI and ML?
Data is used for training AI and ML models.

3. What are the types of data?
Numerical data, Categorical, text, image and audio data

4. What is the difference between structured and unstructured data?
Structure data is data which can be organised in row and column.
Eg: excel sheet
Unstructured data is data which cannot be stored in row and columns.
Eg: image or video

5. Give some real-life examples of data.

Student Name	Study Hours	Attendance	Marks
Rahul	3	90%	85/100
Priya	2	80%	70/100
Aman	5	95%	92/100

Assignment - 45

1. What is Artificial Intelligence?
Artificial Intelligence 'Ai' mean a system which behaves like human intelligence. It mean making a computer or machine think and work like humans.

2. Where is AI used in daily life?
Mobile voice assistance like Siri and google assistant, YouTube and Netflix recommendation system, self-driving car, online shopping apps, Face unlock in smartphones, and chatbots & customers support.

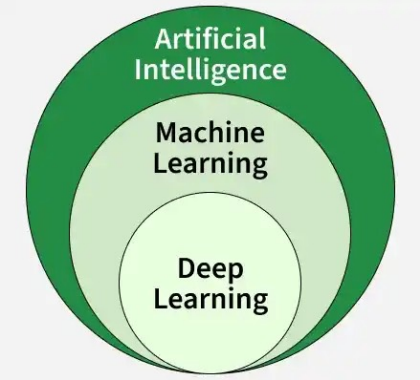
3. What are the advantages of AI?
Fast work, less human error, smart automation, better customer support and helps in data analysis.

4. What is the difference between AI and humans?
Ai works fast , does not get tired, learns from data and no real emotional.
Human brain thinks emotionally, need rest, learns from experience and has emotions and feelings.

5. Name some popular AI tools.
ChatGPT, Google Assistant, Alexa, Siri, Grammarly, microAi and Google Translate.

Assignment - 46

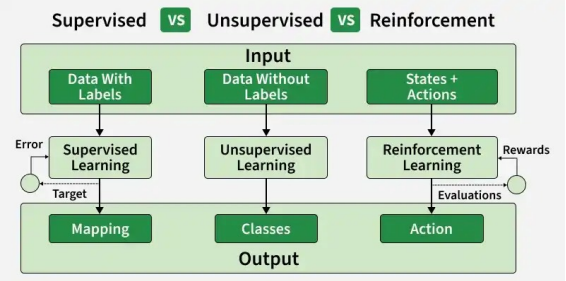
1. What is Machine Learning?
Machine learning (ML) is a part of Artificial Intelligence (AI). It allows computer to learn from data without being fully programmed, ML systems improve automatically by analysing example and patterns.



2. How is ML different from AI?
Artificial Intelligence is the overarching vision, while machine learning is the primary toolkit we use to achieve it. In simple words, AI is a goal and ML is a the method to achieve it.

3. Name some real-life uses of ML.
Netflix recommended system, Gmail spam email detection, Google Maps traffic prediction, Amazon recommending products and Mobile face unlock system.

4. What are the types of Machine Learning?
There are 3 types of machine learning:
Supervised learning: Learning using labeled data
Unsupervised learning: Finding hidden patterns in data
Reinforcement Learning: learning using reward and punishment



5. Why is data important in ML?
Data is very important because ML learning using data.

Assignment - 47

1. Give some real-life examples of AI and ML.
AI and ML is used in day today life such as health care, phone, shopping app, etc.
Example: voice assistant, camera filter, smart watch track heart rate, product suggestion system, etc.

2. How does YouTube use Machine Learning?
It uses ML for suggesting user's favourite videos, that suggestion system is categorise into unsupervised learning.

3. How is AI helping in healthcare?
Ai helps doctor to detect disease faster.
Smart watch tracks heart rate, steps and sleep
Hospital use AI for patient report analysis.
ML is predict health risk from patient's data.
Robots assist in surgeries in advanced hospital

4. What are the advantages of AI in daily life?
It makes works faster, increasing accuracy, personalise experience, save time, and smart automation.

5. Name some popular AI tools used in 2026.
ChatGPT, Google Gemini, Microsoft copilot, Grammaly AI, Canva AI, Midjourney, Github Copilot.

Assignment - 48

1.What is the purpose of the LIMIT clause in SQL?
It restricts the number of rows returned by query.
Syntax:

```
SELECT Studname FROM Student  LIMIT 5;
```

2.How is LIMIT different from ORDER BY?
LIMIT is commonly used with ORDER BY to get top n record.
Eg: query to get top 5 student who have scored highest.

```
SELECT * FROM Student ORDER BY StudMarks DESC LIMIT 5;
```

3.How can you find the Top 3 highest-rated movies using LIMIT?
We can find it using ORDER BY.
Syntax:

```
SELECT * FROM movie ORDER BY imdb_rating ASC LIMIT 3;
```

4.What does the query LIMIT 2,3 do?
It helps us to skip some records and display the next row.
Syntax:

```
SELECT * FROM movie LIMIT 2,3;
```

Output:

movie_id	movie_name
3	Dangal
4	KGF 2
5	3 Idiots

5.What will be the output of the following query?

```
SELECT movie_name, imdb_rating
FROM Movies
ORDER BY imdb_rating DESC
LIMIT 2;
```

F

movie_name	imdb_rating
Kantara	8.7
Pushpa 2	8.5

Assignment - 49

1.What are Logical Operators in SQL and why are they used?
Logical Operators are used to combine multiple conditions in a sql query.
The three main logical operators are:
AND, OR, NOT
These operators are commonly used with the WHERE clause to filter records.

2.What is the difference between AND and OR operators?
AND operator is true when both conditions are true. OR operator is true when anyone condition is true.

3.What does the NOT operator do in SQL?
NOT is used to exclude records matching a condition

4.What will SQL execute first in this condition?

```
WHERE movie_language = 'Hindi'
OR movie_language = 'Kannada'
AND imdb_rating > 8
```

AND condition will be executed first.

5.What is the output of the following query?

```
SELECT *
FROM Movies
WHERE NOT genre = 'Comedy'
AND imdb_rating > 8;
```

All movies will be displayed who's rating is above 8 and genre is not comedy

Assignment - 50

1. What is the purpose of ORDER BY in SQL?
ORDER BY is used to sort data in ascending or descending order.
Syntax:
```
SELECT column_name FROM table_name ORDER BY column_name;
```

2. What is the difference between ASC and DESC?
ASC means Ascending Order & DESC means Descending Order.

3. What happens if ASC or DESC is not specified in ORDER BY?
If ASC or DESC is not specified in ORDER BY then it data will be sorted by ASC (ascending order).

4. Can ORDER BY be used on multiple columns? Explain with an example.
We can sort data using more than one column.
For example:
```
SELECT movie_name, movie_language, imdb_rating FROM Movies ORDER BY
movie_language ASC, imdb_rating DESC;
```
First SQL sorts by language, then within each language, it sorts by rating.

5. What will be the output of this query?

```
SELECT movie_name, imdb_rating
FROM Movies
ORDER BY imdb_rating DESC;
```

It will display all the movie name and imdb_rating in descending order.

Assignment - 51

1.What is the purpose of the UPDATE statement in SQL?
The UPDATE is used to modify the existing records in a table. It allows us change one or more values for selected rows.

2.Why should we use the WHERE clause with UPDATE?
WHERE is used to select the particular row and value. If we don't used WHERE is not used, all row will be updated.

3.What happens if an UPDATE statement is executed without a WHERE clause?
If we don't used Where is not used, all rows will be updated.

4.How can you update multiple columns in a single UPDATE statement?
To update multiple columns at once, you use a single SET clause and separate each column-value pair with a comma (,).

5.What is the safer way to update a record: using movie_name or movie_id? Why?
Using movie_id is much safer than using movie_name.
A movie_id is typically designed as a Primary Key, meaning it is completely unique to every single record. On the other hand, movie_name is not guaranteed to be unique.

Assignment - 52

1.What are Aggregate Functions in SQL?
Aggregate Function perform calculations on multiple rows and return a single result.
They are commonly used for data analysis, reporting, and business insights.
The most common used Aggregate functions are:
COUNT()
SUM()
AVG()
MAX()
MIN()

2.What is the difference between COUNT(*) and COUNT(column_name)?
COUNT(*) shows total number of rows in whole table and COUNT(column_name) shows number of rows in a particular column.

3.How does AVG() work in SQL?
AVG() is used to calculate the average value.

4.Which Aggregate Function would you use to find the highest box office collection?
We can use MAX() to find highest box office collections.

5.What will be the output of the following query?

```
SELECT COUNT(*),
AVG(imdb_rating),
MAX(imdb_rating)
FROM Movies;
```

It will shows total no of rows in whole table, average of imdb_rating, and maximum value of imdb_rating.

Assignment - 53

1.What are Date and Time Functions in SQL?
Dates and times functions are used to work with dates and times on SQL.
It helps us:
Get the current date
Get the current time
Extract year, month, and day
Calculate date differences
Generate reports based on dates

2.What is the difference between CURRENT_DATE() and NOW()?
CURRENT_DATE() returns only date and NOW() returns both date and time.

3.How can you extract the year from a date column?
We can used YEAR () function to extract only year.

4.What is the purpose of the DATEDIFF() function?
It returns the difference between the two dates.

5.What will be the output of the following query?

```
SELECT YEAR('2024-12-05');
```

The output will be "2024"

Assignment - 54

1.What is the purpose of the DELETE statement in SQL?
The DELECT is used to remove existing records (rows) from a table.
It deletes data from the table but does not the whole table.
Syntax:

```
DELECT FROM table_name WHERE conditions;
```

2.What is the difference between DELETE and DROP?
DELECT deletes the rows from table, DROP deletes the whole table.

3.What happens if DELETE is executed without a WHERE clause?
It will delete the every row in the table.

4.Why is using movie_id in a DELETE statement safer than using movie_name?
movie_id is much safer because 2 different movies can have same name that will lead to deletion of both but they will unique id.

5.What will be the result of the following query?

```
DELETE FROM Movies
WHERE imdb_rating < 8;
```

It will delete all the records whose imdb_rating is less than 8.

Assignment - 55

1.What is a Foreign Key in SQL?
A foreign Key is a column in one table that refers to the primary key of another table.

2.What is the difference between a Primary Key and a Foreign Key?
A primary key is a column that uniquely identifies each record in a table.
A foreign key is a primary key or column imported from another table.

3.Why are Foreign Keys important in a database?

Why Foreign Keys are important: They enforce referential integrity, keeping data consistent across related tables and preventing orphan records.

4.What happens when you insert a Foreign Key value that does not exist in the Parent Table?

Inserting a non-existent value: The database rejects the insertion with a foreign key constraint violation error to protect data validity.

5.Explain the relationship between Movies and Reviews tables using Primary Key and Foreign Key.

Movies and Reviews relationship: The `Movies` table has a Primary Key (like `movie_id`), while the `Reviews` table contains a Foreign Key (`movie_id`) pointing to it, establishing a one-to-many relationship where one movie can have multiple reviews.

Assignment - 56

1.**What is a Primary Key in SQL?**
It is a column (or a set of columns) used to uniquely identify each row in a database table. Every table should ideally have a primary key, and no two rows can have the same primary key value.

2.**What are the rules of a Primary Key?**

- **Uniqueness:** Every value in the primary key column must be distinct.
- **Not Null:** It cannot contain any `NULL` values.
- **Single per Table:** A table can have only **one** primary key (though that key can consist of multiple columns).

3.**Why can't a Primary Key contain NULL values?**
Because `NULL` represents an unknown or missing value. Since a primary key's sole purpose is to reliably and uniquely identify every single row, allowing `NULL` would mean having rows that cannot be uniquely identified.

4.**What is a Composite Primary Key?**
A primary key made by combining **two or more columns** together. This is used when a single column isn't enough to guarantee uniqueness, but the combination of columns is entirely unique (for example, combining `student_id` and `course_id` in an enrollment table).

5.**What will happen if you try to insert a duplicate Primary Key value?**
The database management system will reject the insertion and throw a **primary key constraint violation error**, protecting the table from duplicate data and preserving row-level uniqueness.

Assignment - 57

1.**What are String Functions in SQL?** Built-in functions used to manipulate, format, search, or extract information from text data (strings) stored in database tables. They allow you to clean up data or change how text is displayed in your query results.

2.**What is the difference between UPPER() and LOWER()?**
UPPER(column) converts all letters in a text string to uppercase (e.g., 'inception' becomes 'INCEPTION').
LOWER(column) converts all letters in a text string to lowercase (e.g., 'INCEPTION' becomes 'inception').

3.**How does the CONCAT() function work?** It joins two or more text strings or column values together into a single combined string. If you pass multiple arguments separated by commas, it links them end-to-end in that exact order.

4.**What is the purpose of the TRIM() function?**
It removes leading and trailing whitespace (extra spaces at the beginning or end) from a text string, which is useful for cleaning up messy user inputs or data import errors.

5.**What will be the output of the following query?**
SQL

```
SELECT CONCAT(UPPER(movie_name), ' - ', movie_language)FROM Movies;
```

Output description: For every row in the Movies table, this query will return a single combined string column where the movie name is converted entirely to uppercase letters, followed by a hyphen separator (-), followed by the movie language in its original casing (e.g., INCEPTION - English).

Assignment - 58

1.What are Table Relationships in SQL?
Table Relationships are used to connect two or more tables in a database.
Relationship are usually created using :
Primary key(PK)
Foreign key(FK)

2.What is the difference between One-to-One and One-to-Many relationships?
One record in table A is related to only one record in table B.
Example:
A movie has one censor certificate.
One record in table A can have many related record in table B.
Example:
One movie can have many reviews.

3.Why is a Foreign Key important in creating relationships?
Because help organise data efficiently and reduce data duplication.

4.What is a Many-to-Many relationship and how is it implemented?
Many records in Table A can be related to many records in table B.
It is implemented by using junction table 9
Example:
Movie and Actors

5.Explain the relationship between Movies and Reviews tables.
It has one to many relationship because one movie can have many reviews.

Assignment - 59

1. What is the purpose of the UPDATE statement in SQL?
The Update statement is used to modify existing records in a table.

2. Why should we use the WHERE clause with UPDATE?
We should use WHERE clause with UPDATE to modify the a particular row.

3. What happens if an UPDATE statement is executed without a WHERE clause?
If don't use WHERE clause it will update all the rows.

4. How can you update multiple columns in a single UPDATE statement?
Yes we can update multiple columns in a single UPDATE statement.

5. What is the safer way to update a record: using movie_name or movie_id? Why?
movie_id is safer way to update a record because 2 different movies can have same name but they will have unique id.

Assignment - 60

1.What is the purpose of a JOIN in SQL?
SQL joins are used to combine data from two or more tables based on a related column.

2.What is the difference between INNER JOIN and LEFT JOIN?
INNER JOIN returns only the matching records from both tables.
LEFT JOIN returns all records from the left table and matching records from the right table.

3.How does RIGHT JOIN work?
RIGHT JOIN returns all records from the right table and matching records from the left table.

4.When would you use FULL JOIN?
When we want to join both the tables, matching records are combined and non-matching records show NULL values.

5.How can you find unmatched records using JOINs?
We can use FULL JOIN to find unmatched records.

Assignment - 61

1.What is a Subquery in SQL?
A Subquery is a query written inside sql query. It is also called an Inner query or Nested Query.
The result of the inner query is used by the outer query.

2.What is the difference between a Subquery and a Normal Query?
Normal Query consist of a single standalone SQL statement that directly retrieves, modifies or filter data from tables in a single step.
Subquery involves two or more queries working together in a parent-child relationships. The outer query relies on the result returned by the inner subquery to process its own logic.

3.What is the difference between IN and EXISTS in Subqueries?
IN operators : Compares a specific column value against a literal list or set of values returned by the inner query. It processes the full subquery first before matching values .
EXISTS operator: Checks for the existence of the rows that satisfy a conditions inside the subquery. It evaluates to trues as soon as it finds the first matching records and stops searching immediately, making it significantly more efficient when working with large subquery datasets.

4.What are correlated subqueries?
A Correlated subquery is a type of nested subquery that depends on values from the outer query for its execution.
Unlike a standard subquery (which runs once independently), a correlated subquery executed repeatedly - once for every individual row evaluated by the outer query. It references columns form the outer table to perform row by row calculations or comparisons.

5.Can a Subquery be used in SELECT, FROM, and WHERE clauses?
Yes, a subquery can e flexible used across all three clauses depending on the requirements:
In WHERE Clause: Used as a filtering condition to evaluate against values returned by (most common use case).
In SELECT Clause: Used as a temporary table (often called a derived table or return a single value alongside main table columns.
In FROM Clause: Used as a temporary table (often called a derived table or inline view) that the outer query select data from.

Assignment - 62

1.What is a View in SQL?
A view is a virtual table created from one or more SQL queries.

2.How is a View different from a Table?
A view does not store data physically. Instead, it stored the SQL query and displays data from the original table whenever it is accessed.

3.What are the advantages of using Views?
Simplifies complex queries.
Improve data security
Hides sensitives columns
Reusable query logic
Easier reporting

4.How do you create and delete a View?
Creating view:
CREATE VIEW view_name AS
SELECT columns
FROM table_name;

Delete a view:
DROP VIEW view_name;

5.Why are Views commonly used in reporting systems?
It stored the SQL query and displays data from the original table whenever it is accessed.

Assignment - 63

1.What are Window Functions in SQL?
Window Function perform calculations across a set of rows related to the current row.

2.What is the difference between GROUP BY and Window Functions?
The GROUP BY reduces the number the number of rows, Window Function do not reduce the number of rows.

3.What is the difference between RANK(), DENSE_RANK(), and ROW_NUMBER()?
RANK() gives ranking to rows .
DENSE_RANK() is similar to RANK() but does not skip rank numbers.
ROW_NUMBER() assigns a unique number to each row.

4.How does PARTITION BY work in Window Functions?
PARTITION BY divides data into groups before applying the window function.

5.What are the uses of LAG() and LEAD() functions?
LAG() accesses data from the previous row.
LEAD() accesses data from the next row

Assignment - 64

1.What is a window function in SQL?
A window function is used to perform calculations across a set of rows related to the current row.

2.Why do we use OVER() in SQL?
OVER() defines the window for calculation. It tells SQL that the function should work over a group of rows.

3.What is the difference between GROUP BY and window functions?

GROUP BY	WINDOW FUNCTION
Combines the row	Does not combine rows
Output has fewer rows	Returns original rows with calculation
Used for summary reports	Used for row - wise analysis
Cannot easily show detail rows with summary	Can show detail rows with summary

4.What is the difference between ROW_NUMBER() and RANK()?
ROW_NUMBER() gives a unique number to every row.
RANK() gives same rank for same values and skips rank

5.What is the difference between RANK() and DENSE_RANK()?
RANK() gives same rank for rank for same values and skips rank
DENSE_RANK() gives same rank for same values but does not skip rank

6.What is the use of PARTITION BY?
PARTITION BY devices rows into groups.

7.How can you find department-wise employee ranking?
We can use PARTITION BY and RANK().
PARTITION BY is used to divide employees their department wise and RANK() is used to rank them

```
SELECT emp_name, department, salary,
RANK() OVER(
       PARTITION BY department
       ORDER BY salary DESC)
AS dept_rank FROM Employees;
```

8.What is the use of LAG()?
LAG() return the previous row value. It is mostly used to compare the current row with the previous row.

9.What is the use of LEAD()?
LEAD() return the next row value. It is mostly used to compares the current row with the next row.

10.How can you calculate sales growth using SQL?
We can calculate using sales growth using LEAD() and LAG().

11.What is the use of NTILE()?
NTILE(n) divides rows into n groups or buckets.

12.What is the use of CUME_DIST()?
CUME_DIST() means cumulative distribution. It returns the relative position of a row within the result set.

13.What is a running total?
A running total means a cumulative total that increases row by row.

14.How can you calculate a running total using a window function?

```
SELECT month_name, sales,
       SUM(sales) OVER(
           ORDER BY month_no
           ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW
       ) AS running_total
FROM SalesData;
```

15.Why is LAST_VALUE() sometimes confusing?
LAST_VALUE() usually needs a proper window frame, otherwise it may not return the expected final value

Assignment - 65

1.What is CASE Statement in SQL?
CASE Statement is used to apply condition in SQL query. It works like `if-else` condition

```
CASE
    WHEN condition THEN result
    WHEN condition THEN result
    ELSE result
END
```

2.Why do we use CASE Statement?
CASE Statement is used when we want conditions output in SQL.

3.What is the syntax of CASE Statement?

```
CASE
    WHEN condition THEN result
    WHEN condition THEN result
    ELSE result
END
```

4.What is the use of ELSE in CASE Statement?
ELSE return default result if above conditions are not true.

5.Tricky Question: What happens if no WHEN condition is true and ELSE is not used?
Then SQL will return NULL.

Assignment - 66

1.What is Index in SQL?
An Index is used to search data faster from a table. It works like the index page of a book.

2.Why do we use Index?
A table has many records, index helps the database find required data quickly.

3.What is the syntax to create an Index?

```
CREATE INDEX idx_name
ON employees(name);
```

4.What is Composite Index?
Composite Index means index created on more than one column.

5.Tricky Question: Why should we not create Index on every column?
Because column is rarely searched.

Assignment - 67

1.What is Query Optimization in SQL?
Query Optimisation means writing SQL queries in a better way so that data can be retrieved faster.

2.Why should we avoid `SELECT *`?
Because it can make query slow when table has many columns.

3.How does Index help in Query Optimization?
It is useful on columns used in WHERE, JOIN, ORDER BY, and GROUP BY.

4.What is the use of EXPLAIN in SQL?
EXPLAIN is used to check how SQL query is executed.

5.Tricky Question: Can too many indexes slow down a database?
If we use functions on indexed columns, sometimes index may not work properly.

Assignment - 68

1.What is Stored Procedure in SQL?
A Stored Procedure is a saved SQL program inside the database.

2.Why do we use Stored Procedure?
Stored Procedure is used to save time and avoid writing the same SQL query again and again,

3.How do we execute a Stored Procedure?
To run a stored procedure, we use CALL

4.What is the use of parameter in Stored Procedure?
A parameter is used to pass value inside a stored procedure.

```
DELIMITER //

CREATE PROCEDURE get_employee_by_department(IN dept_name VARCHAR(100))
BEGIN
    SELECT * FROM employees
    WHERE department = dept_name;
END //

DELIMITER ;
```

5.Tricky Question: What is the difference between Stored Procedure and Normal SQL Query?

Normal Query	Stored procedure
Written every time	Created once and reused
Not saved permanently	Saved inside database
Good for small work	Good for repeated work
Executed directly	Executed using CALL

Assignment - 69

1.What is Trigger in SQL?
A trigger is a special SQL program that runs automatically when any action happens in a table.

2.Why do we use Trigger?
Trigger is used to automate database work

3.What is the difference between BEFORE Trigger and AFTER Trigger?
AFTER runs after action and BEFORE runs before the action.

4.What is the use of OLD and NEW in Trigger?
OLD and NEW are used to access values inside trigger.

5.Tricky Question: Can we use OLD and NEW both in every Trigger?
Yes we use OLD and NEW both in every Trigger.

Assignment - 70

1.What is the use of `st.number_input()` in a Streamlit calculator app?
It allows users to give numerical input.

2.Why do we use `st.selectbox()` for choosing an operation?
It allow users to select athematic operations.

3.Why should we check whether the second number is zero before division?
Because division by zero is not allowed in mathematics.

4.What will happen if we write calculator logic outside the `st.button()` block?
Calculation will not be calculate and result will not be printed when we press calculate button.

5.Why is `streamlit run app.py` preferred for running a Streamlit app instead of only `python app.py`?
Because app wont get started by python app.py

Assignment - 71

1.What is the first step in setting up a Streamlit project?
To check python is installed or not. Then create project folder.And to create virtual environment.

2.Why do we create a virtual environment before installing Streamlit?
We create a virtual environment before installing streamlit to avoid package conflicts between different projects.

3.What is the difference between `python –m venv venv` and `pip install streamlit`?
”python -m venv venv” is used to create virtual environment. And "pip install stramlit” is used to install streamlit packages.

4.Why should we activate the virtual environment before installing packages?
In Windows:
venv/Scripts/activate
In MacBook:
source venv/bin/activate

5.If `streamlit run app.py` shows an error, what basic things should we check first?

Error	**Reason**	**Solution**
python is not recognized	Python path is not set	Install Python properly and add it to PATH
pip is not recognized	pip is not available	Reinstall Python with pip option
streamlit is not recognized	Streamlit is not installed	Run pip install streamlit
code is not recognized	VS Code command is not enabled	Install VS Code or enable code command
app.py not found	CMD is in the wrong folder	Open CMD inside the correct project folder

Assignment - 72

1.What is Streamlit?
Streamlit is a python framework used to create web applications, dashboard, data science apps, machine learning apps, machine learning, and AI project interface.

2.Why is Streamlit useful for Python developers?
It helps developer convert Python code into a web app without writing separate HTML, CSS, and JavaScript for basic applications.

3.What type of applications can we create using Streamlit?
Streamlit is mainly used to create Python-based web applications.
We can create:

- Sales dashboard
- Student result app
- Weather data app
- Machine Learning prediction app
- Resume analyser
- Chatbot interface
- Data analysis dashboard

4.Can Streamlit create frontend and backend using only Python? Explain with an example.
In normal web development, frontend and backend are usually created separately.
But in streamlit, we can create both using python.
Formula:
Frontend = Buttons, text boxes, sliders, tables, charts
Backend = Python logic, calculations, data processing
For Example:

```
import streamlit as st

price = st.number_input(”Enter production price”, 1, 100000)
quantity = st.number_input(”Enter quantity:”, 1, 100)

total = price * quantity

st.write(”total Amount:”, total)
```

5.If a company wants a quick dashboard from Excel or CSV data, why can Streamlit be a good choice?
Streamlit is a great choice because it let you turn Excel or CSV files into interactive, web-based dashboard in minutes using just standard python with no frontend web-development required.

Assignment - 73

1.What is a pie chart?
A pie chart is a circular chart used to show how a full quantity is divided into different parts.

2.Why do we use `px.pie()` in a Streamlit app?
We use `px.pie()`to create a pie chart, from ploty Express.

3.What is the difference between the `values` and `names` parameters in `px.pie()`?
Values is for numeric column for slice size and names is for category column for slice names.

4.What will happen if the column name passed in `values` does not exist in the DataFrame?
If the column name passed in passed in values does not exist in the DataFrame will give you ValueError.

5.Why is a pie chart not a good choice when the data has too many categories?
A pie chart becomes a cluttered, unreadable rainbow of tiny slices when you have too many categories, making it impossible to compare sizes or tell which data point is actually bigger.
(Sent it already)

Assignment - 74

1.What is Data Preprocessing?
Data Preprocessing is the process of cleaning and preparing raw data before giving it to a Machine Learning model.

2.Why is Data Preprocessing important in Machine Learning?
If data is incorrect, incomplete, or messy, the model can give wrong predictions.

3.What are missing values?
Missing value means some data is empty or not available.

4.What are duplicate values?
Duplicate values mean the same row is repeated more than one time.

5.Why do we convert text data into numbers?
Computers and machine learning algorithms cannot directly read, understand, or process words—they only speak the language of math.

Assignment - 75

1.What is Model Comparison?
Model Comparison means comparing two or more Machine Learning models to find which model is performing better.
In Machine Learning, we do not depend on only one model. We train multiple model and compare their result.
For example:
Linear Regression vs Decision Tree Regression

2.Why do we compare Machine Learning models?
We compare model to select the best model for prediction. One model may perform better on one dataset, and another model may perform better on another dataset.
For Example: For house price prediction:
Linear Regression may work well for simple linear data.
Decision Tree may work well for non-linear data.

3.What is the difference between Linear Regression and Decision Tree Regression?

Linear Regression	**Decision tree Regression**
Uses straight-line relationship	Uses decision rules
Good for linear data	Good for non-linear data
Easy to understand	Easy to visualise
May underfit complex data	May overfit small data
Predict continuous values	Predicts continuous values

4.What is RMSE?
RMSE is Square root of MSE

5.How do we choose the best model?
MAE is lower is better
MSE lower is better
RMSE lower is better
R2 Score higher is better
If Decision Tree has lower RMSE and higher R2 Score, then Decision Tree is better.

Assignment - 76

1.What is Pandas?
Pandas is a python library used to handle data before using it in Machine Learning. It helps us load data, view data, understand data, and select input and output columns.
Example:
Pandas = Data handling tool
ML Model = Learning Tool

2.Why do we use Pandas in Machine Learning?
It is used for because Machine Learning needs data in a proper table format

3.What is a DataFrame?
A DataFrame is a table-like structure in pandas.

4.What is the difference between X and y?
Usually X is used to store input data.

5.Which command is used to read a CSV file?
pd.read_csv()

Assignment - 77

1.What is Pickle in Python?
Pickle is a python module used save python objects into a file.

2.What is a `.pkl` file?
A .pkl file is a file where Python objects are saved using the pickle module

3.Why do we save ML models using Pickle?
We save ML models using pickle because we dont need to train the model again and again.

4.What is the use of `pickle.dump()`?
It is used save the data in pickle.

5.What is the use of `pickle.load()`?
It is used to load the data from pickle.

Assignment - 78

1.What is Time Series Forecasting?
Time Series Forecasting is a Technique used to predict future values using past data arranged in time order.

2.What is Time Series Data?
Time Series Data is data collected according to time.

3.Why is time important in Time Series Forecasting?
Time is important because to see the how data changes overtime, so that model can find hidden patterns and predict accordingly.

4.Give two real-life examples of Time Series Forecasting.
Weather prediction
Sales forecasting

5.What is the simple formula for forecasting the next value?
Next value = Last value + Difference

Assignment - 80

1.What are the main steps in a Machine Learning project?

2.Problem Definition
3.Data collection
4.Import Required Libraries
5.Load DataSet
6.Convert Data into DataFrame
7.EDA
8.Data Cleaning
9.Feature Engineering
10.Feature Selection
11.Feature scaling
12.Define X and Y
13.Train Test Split
14.Algorithm Selection
15.Model Training
16.Prediction
17.Evaluation
18.Hyper-parameter Tuning
19.Cross Validation
20.Model Optimisation
21.Model Comparison
22.Select Best Model
23.Save Trained Model
24.Deploy Model
25.Monitor, Retrain and Update Model

2.Why is EDA important before model training?
To understand the data so that we can perform Data cleaning. This helps ml model to be accurate.

3.What is the difference between X and Y in Machine Learning?
X and Y means target variable.
X is usually used for input columns(Features)
Y is usually used for output columns(Final Marks)
For example:
Features: Study Hours, Attendance
Target: Final Marks

4.Why do we use train test split?
Train test split means dividing data into data and testing data. This helps check whether the model works well on new data or not.

5.Why should we save and deploy a trained model?
We save it so that we can use it later without training it again

Assignment - 81

1.What is feature scaling in Machine Learning?
Feature scaling in the process of bringing numerical features into a similar range.
In machine Learning, some columns may have small values and some columns may have very large values.
Feature scaling helps balance these values.

2.Why is feature scaling important for distance-based algorithms?
Feature scaling is important for distance-based algorithms because for example KNN checks distance between data points. If one column has very large values, distance calculation can become biased.
So feature scaling is very useful for distance-based algorithms.

3.What is the difference between standardization and normalization?
Standardisation converts value around mean 0 and standard deviation 1.
Normalisation converts value to fix range usually 0 to 1.

4.Why should we fit scaler only on training data?
We fit the scaler only on training data to prevent data leakage, ensuring the test set remains an unbiased, completely unseen representation of real-world data.

5.Which Machine Learning algorithms usually need feature scaling?
Linear Regression, Logistic Regression, KNN, SVM, K-Means, Neural Network, PCA

Assignment - 82

1.What is feature engineering in Machine Learning?
Feature Engineering is the process of the process of creating, selecting, or improving input columns for a machine learning model.

2.Why is feature engineering important for model performance?
It is important because good features helps the model learn better patterns. Even a strong algorithm may perform poorly if the features are not useful.

3.What is the difference between feature and target?
Feature is input columns used by model .
Target is output column model predicts.

4.What is feature selection?
Feature Selection means choosing only useful input columns for the model.
Not every columns is useful. Some columns may confuse the model or increase training time.

5.Why do we encode categorical features before model training?
ML model usually need numbers, so categories must be converted into numbers.

Assignment - 83

1.What is KNN in Machine Learning?
KNN is a Machine Learning algorithm used for classification and regression problems.

2.How does KNN make predictions?
It predicts the result of a new data point by checking the nearest data point around it.

3.What is the role of K in KNN?
The value of K tells how many nearest neighbour will be checked.

4.Why is feature scaling important in KNN?
Feature Scaling is very important in KNN because KNN uses distance. If one feature has very large values, it can dominate the distance calculation.

5.What is the difference between KNN classification and KNN regression?
KNN classification is used to predict a category or class.
KNN regression is used to predict numerical value.

Assignment - 84

1.What is SVM in Machine Learning?
SVM is a supervised Machine Learning algorithm. It learns from labelled data and mainly used for classification problems. But it is also used for regression

2.Why is it called Support Vector Machine?
SVM is called Support Vector Machine because it uses special data points called support vectors.

3.What are support vectors?
Support vector are the special data points that are closest to the decision boundary.

4.What is the difference between Linear SVM and Non-Linear SVM?
Linear SVM is used when data can be separated using a straight line.
Non-linear SVM is used when data cannot be separated by a straight line.

5.What is RBF kernel in SVM?
RBF stands for Radial Basis Function. RBF is a popular kernel used in SVM for non - linear data.

Assignment - 85

1.What is a Weather API, and how is it used in Python?
A Weather API is a service that provides weather information throughout the internet.

2.What is the difference between an API endpoint and an API key?
An API Endpoint mean the URL used to access an API.
An API key is a unique code used to identify and verify a user while accessing an API.

3.How can you access weather data without using an API key?
There are many free websites which provides data without API key. For example Open-meteo.
Which provides free url which can be used to access the data.
For example:

```
import requests

url = "https://api.open-meteo.com/v1/forecast?
latitude=28.61&longitude=77.20&current=temperature_2m&timezone=auto"

response = requests.get(url)

data = response.json()

temperature = data["current"]["temperature_2m"]

print("Current Temperature:", temperature, "°C")
```

4.Why should the status code be checked before reading API data?
200 - request successful
400 - Incorrect request
404 - Resource not found
500 - Server error

5.What will happen if the city exists in different countries with the same name?
If multiple cities share the same name, the API will either default to the most prominent one, return a list of matching locations, or serve weather data for the wrong city unless you specify coordinates or country codes.

Assignment - 86

1.What is a Classification Report?
A Classification Report is used check the performance of a classification model.

2.What is the difference between Precision and recall?

Precision	Recall
Starts from Predicted Positive cases	Starts from Actual Positive cases
Checks how many Positive predictions were correct	Checks how many actual Positive cases were found
Focuses on False Positive	Focuses on False Negative
Formula: TP / (TP + FP)	Formula: TP / (TP + FN)

3.Why can accuracy give the wrong impression for an imbalanced dataset?
Accuracy can give the wrong impression when the dataset is imbalance. Suppose there are 100 transaction. Because the model predicts every transaction as Genuine. The model correctly predicts all 95 Genuine transactions.The model has 95% accuracy, but it did not find even one Fraud transaction.
Therefore, we should not check only accuracy.We should also check Precision, Recall and F1-Score

4.Can the Weighted Average be high when the model performs poorly for a smaller class?
Yes, the weighted average can remain high because performance on the majority class dominates the calculation, easily masking poor performance on smaller classes.

5.Which metric should be checked when missing an actual Positive case is dangerous?
Recall (or Sensitivity), because it directly measures how many actual positive cases were correctly identified and penalizes false negatives.

Assignment - 87

1.What is Streamlit App Deployment?
Streamlit App Deployment means publishing a streamlit application on the internet.

2.Why is the `requirements.txt` file required during deployment?
The requirement.txt file contains the names and versions of python packages used in the project.
Streamlit Cloud reads this file and installs the required packages before running the application.

3.What will happen if a required Python package is missing from `requirements.txt`?
It wont get deployed properly.

4.Why should the `venv` folder not be uploaded to GitHub?
The venv folder contains large, machine specify files that bloat your repository and cause cross-platforms issues, so collaborators should just recreate the environment using requirement.txt file.

5.How does a deployed Streamlit app update after changes are pushed to GitHub?
Signup on streamlit Cloud
Connect GitHub Repository
Select app.py
Deploy the application
Share the application Link

Assignment - 88

1.What is Hyperparameter Tuning?
Hyperparameter tuning is the process of finding the best settings for a Machine Learning model.

2.What is the difference between a parameter and a hyperparameter?

Parameter	Hyperparameter
Learned automatically by the model	Set before model training
Learned from training data	Selected bit the developer
Example: coefficient	Example: max_depth
Changes during training	Remains during one training process

3.Why is Hyperparameter Tuning important?
Hyperparameter are set before model training.
The model does not learn hyper parameters automatically
Hyperparameter tuning helps us find better model settings.

4.What is the difference between Grid Search and Random Search?

Grid Search	Random Search
Tests every given combination	Tests random combinations
Takes more time	Usually takes less time
Good for fewer values	Good for many values
Gives a complete search	Does not test every combination

5.Can incorrect hyperparameter values cause overfitting or underfitting?
Yes, it can cause overfitting or underfitting.

Assignment - 89

1.What is Lasso Regression?
Lasso Regression is a Machine Learning algorithm used for regression problems. It is an advanced form of linear Regression. Lasso Regression predicts continuous values such as price, salary, sales, marks or profit.

2.Why do we use Lasso Regression?
Lasso Regression is used when we want to reduce unnecessary features from the dataset. It can make some feature value exactly zero. This means Lasso can automatically remove less important features.

3.What is L1 regularization?
L1 Regularization adds penalty to the model. This penalty helps reduce the effect of less important features.

4.What is alpha in Lasso Regression?
Alpha is a value that controls the strength of regularization.

5.What is the difference between Linear Regression and Lasso Regression?

Linear Regression	**Lasso Regression**
Uses all features	Can remove less important features
No regularization	Uses L1 regularization
May overfit on complex data	Helps reduce overfitting
Simple regression model	Improved regression model

Assignment - 90

1.What is NLP?
NLP stands Natural Language Processing. It is a part of Artificial Intelligence that helps computers understand, read, write, and respond to human language.

2.Why is NLP important in Artificial Intelligence?
It is important because it is a part of AI to understand human language.

3.What is tokenization in NLP?
Tokenization is used to break text into words or sentences.

4.What is the difference between text classification and sentiment analysis?
Text Classification means dividing text into categories.
Sentiment Analysis means finding emotion or opinion.

5.Why is human language difficult for computers to understand?
Because computers will only understand numerical values.

Assignment - 91

1.Give some real-life examples of NLP.
Google translate, voice assistant, spam email Delection, and sentiment Analysis

2.How does a chatbot use NLP?
A chatbot understands user message and gives replies like human conversation.

3.How is NLP used in Google Translate?
Google Trnslate uses NLP to convert one language into another language.
For Example:
English: I live in Delhi.
Hindi: मैं दिल्ली में रहता हूँ।

4.Why is NLP useful in spam email detection?
It checks words, sentence patterns, Links, and message styles to decide whether an email is spam or not.
For Example:
Email: Congratulations! You won a free prize. Click here now.
Result: Spam email detected

5.What is the use of sentiment analysis in business?
Sentiment Analysis it help to understand custom's feedback are positive or negative.

Assignment - 92

1.What are stop words in NLP?
Stop words are common words that appear many times in a sentence but usually do not aadd much meaning for analysis.

2.Why do we remove stop words from text data?
We remove stop words to reduce unnecessary words from text.

3.Give some examples of common stop words.
Is, am, are, the, in, to, and

4.Why should we be careful while removing the word `not`?
Because it can change the whole meaning of the sentence.

5.How does stop words removal help in text classification?
By removal of stop words it classifies which words are necessary or which are unnecessary.

Assignment - 93

1.What is text data in NLP?
Text data means data written in words, sentences, paragraph, reviews, messages, emails, documents or social media posts.

2.Why is text data important in NLP?
Text data is important because most human communication happens through language.

3.What is the difference between structured and unstructured text data?
Structure Text is the text in fixed format, for example name, city, email, etc.
Unstructured Text is the free writing without fixed format, for example review, comment, paragraph, etc.

4.What are common problems found in raw text data?
Text data is not always clean. It may contain spelling mistakes, extra spaces, emojis, punctuation, mixed languages, short forms, and repeated words.

5.Why do we clean text data before using it in NLP?
Text data is not always clean. It may contain spelling mistakes, extra spaces, emojis, punctuation, mixed language, short forms, and repeated words.

Assignment - 94

1.What is tokenization in NLP?
Tokenization is the process of breaking text into smaller parts.

2.Why is tokenization important before text processing?
Because it is difficult for computers to understand a full sentence directly like humans, so tokenization is done first then text processing, so the computer to understand the text better.

3.What is the difference between word tokenization and sentence tokenization?
Word tokenization means breaking a sentence into words and Sentence tokenization means breaking paragraph into sentences.

4.Why can punctuation create problems during tokenization?
Attached punctuation will create two separate token for the same base word for example "end" vs ”end.”, unnecessary inflating vocabulary size, and fragmenting contextual meaning.

5.Why is basic `split()` not always enough for real NLP projects?
Basic split() splits only on whitespaces, failing to separate punctuation, handle contractions, or recognise multi-term as unified term.

Assignment - 95

1.What is Bag of Words in NLP?
Bag if Words is an NLP technique used to convert text into numbers.

2.Why do we convert text into numbers in NLP?
We convert text into numbers in NLP because ML models need numerical data for training.

3.What is vocabulary in Bag of Words?
In the Bag of Words model, vocabulary is the set of all unique words collected from the entire text dataset.

4.What is the main limitation of Bag of Words?
It is ignores word order, Large vocabulary, less meaning understanding.

5.What is the difference between Bag of Words and normal tokenization?

Feature/Aspect	**Normal Tokenization**	**Bag of Words(BOW)**
Primary Goal	Splits raw text into smaller atomic sentences	Converts text into numerical vectors based on word frequencies.
Word Order/ Grammar	Preserves the original sequence and chronological order of words	Completely discards words order, grammar, and context
Output	A list that contain words	A numerical frequency vector or dictionary

Assignment - 96

1.What is lemmatization in NLP?
Lemmatization is a text preprocessing technique used to convert words into their proper base form.

2.Why do we use lemmatization in text preprocessing?
We use lemmatisation to reduce different forms of the same word into one meaningful base word.
It helps NLP models understand that words like playing, played and plays are connected to the word play.

3.What is the difference between stemming and lemmatization?

Point	Stemming	Lemmatization
Meaning	Cuts word endings	Finds dictionary base form
Accuracy	Less accurate	More accurate
Speed	Faster	Slower than stemming
Output	May not be a proper word	Usually proper word
Example	studies → studi	studies → study

4.Why does lemmatization need part of speech sometimes?
Part of speech means the role of a word in a sentence. A word can be a noun, verb, adjective, or adverb.
Lemmatization gives better result when we provide part of speech.
For example:

```
from nltk.stem import WordNetLemmatizer
lemmatizer = WordNetLemmatizer()
print(lemmatizer.lemmatize("running", pos="v"))
print(lemmatizer.lemmatize("better", pos=”a"))
```

Output:
run
good

5.Why is lemmatization useful in chatbot and search engine projects?
Understands user words better and matches related words.

Assignment - 97

1.What is lowercasing in NLP?
Lowercasing text means converting all letter of text into small letters.

2.Why do we convert text into lowercase before NLP processing?
We convert text to lower case to make text consistent.

3.Which Python method is used for lowercasing text?
In python, we use the lower() method to convert text into lowercase.
For example:
`text.lower()`

4.What is the difference between `Python` and `python` before lowercasing?
Python (capitalized) refers to the programming language itself, whereas python (lowercased) refers to the command-line executable used to run scripts.

5.Why should we be careful while lowercasing words like `US` and `Apple`?
US stands for United States but when we convert to lower case us it becomes normal word
Apple mean is tech-company but when it converted to lower case apple is become a fruit.

Assignment - 97

1.What is Named Entity Recognition in NLP?
Named Entity Recognition is an NLP technique used to find important names from text.

2.Why do we use NER in text processing?
We use NER to extract useful information from large text. It helps computers understand which words are important names or special information.

3.What are common entity types in NER?
Some of the common entities types are:
Person- name of the person, eg: Riya, Aman
Location - city, state , country, place; eg: Riyadh, Saudi Arabia, Berlin, Germany
Organisation - company, college, institute; eg: Google, Microsoft, Tesla
Date - Day, month, year; eg: monday, 2026
Time - Time value; eg: 6pm, 10;30 Am
Money - Currency amount; eg:$5000
Percentage - Percent value; eg: 75%, 20%

4.What is the difference between POS tagging and NER?

POS Tagging	Named Entity Recognition
Finds grammar role go the word	Finds important named entities
Its output consist of Noun, verb, adjective	Its output consist of Person, location, organization
Its main focus is on grammar	Its main focus is extract important information
Eg: Delhi -> noun	Eg: Delhi -> Location

5.How is NER useful in resume screening and news analysis?
NER is important for resume screening because it extract important information from the resume like name of person, in which company is worked in, what her/his qualifications?, how many years of experience he/she has, etc

Assignment - 98

1. What is Parts of Speech Tagging in NLP?
Parts of speech Tagging is an NLP technique used to identify the role of each word grammatically in a sentence.

2.Why is POS tagging important in NLP?
We use POS tagging to understand the grammar structure of a sentence. It helps NLP systems understand how words used in a sentence.

3.What is the difference between noun and verb?
Noun means name of person, place or thing. It means specific identity given to a object or animal. Eg: Casper, Delhi, book.
Verb means action words which describe the action. Eg: run, eat, learn, write, etc.

4.How can the same word have different POS tags in different sentences?
Let see that in a example:
”I have a book”
Here, book is a noun.
”Please book a ticket”
Here, book is a verb.

5.What is the difference between tokenization and POS tagging?
Tokenization breaks text into words, its output is tokens, Eg: Riya, is, learning. Use Text splitting
POS Tagging gives grammar label to each word, its output POS tags, Eg: Riya/NNP, is/VBZ, Learning/VBG. Grammar understanding

Assignment - 99

1.What is punctuation removal in NLP?
Punctuation removal means removing symbols like comma, full stop, question mark, exclamation mark, and other special marks from text.

2.Why do we remove punctuation from text data?
We remove function because many NLP tasks focus mainly on words, not symbols. Punctuation can create extra tokens and make text processing less clean.

3.How can punctuation affect tokenization?
Punctuation can create extra tokens and make text processing less clean.

4.Which Python module is useful for punctuation removal?
In Python, we can remove punctuation using the strung module. THe string.punctuation contains common punctation symbols common punctuation symbols.

5.Why should we be careful while removing punctuation from emails and URLs?
Punctuation should not always be removed blindly. Sometimes punctuation changes the meaning of text. For example:
Sentence 1: Let’s eat, Riya.
Sentence 2: Let’s eat Riya.
In the first sentence, comma makes the meaning clear. In the second sentence, meaning becomes different.

Assignment - 100

1.What is stemming in NLP?
Stemming is text preprocessing technique used to reduce words to their root form.
It removes extra word ending like ing, ed, s, and es.

2.Why do we use stemming in text preprocessing?
We use stemming to reduce different forms of the same word into one common form. This helps NLP models treat similar words as related words.

3.What is the difference between stemming and lemmatization?

Point	Stemming	Lemmatization
Meaning	Cuts word endings	Converts word to dictionary base form
Speed	Faster	Slower than stemming
Accuracy	Less accurate	More accurate
Output	May not be a proper word	Usually proper word
Example	studies → studi	studies → study

4.Why can stemming sometimes produce words like `studi`?
Stemming works by cutting the ending part of a word. It does not always care about grammar or dictionary meaning. It mainly follows rules to shorten words.

5.Where is stemming used in real-life NLP projects?

NLP Project	Use of Stemming
Search Engine	Matches similar word forms
Spam Detection	Reduces repeated word variations
Text Classification	Groups similar words
Topic Modeling	Finds common topic words
Information Retrieval	Improves document searching

Assignment - 101

1.What is language detection in NLP?
Language detection is an NLP task used to identify the language of a given text.

2.Why is language detection important in multilingual applications?
We use language. Detection when text can come in different languages. It helps the system close the correct language model, translation tool,

3.How does language detection help chatbots?
Language detection works by checking patterns in text. Different language have different words, characters, and styles.

4.Why is mixed language text difficult for language detection?
Mixed-language text (such as *Hinglish* or code-mixing like *"Aaj ka weather bohot cold hai"*) poses significant challenges for language detection because:
Overlapping Scripts & Vocabularies: Words from different languages appear together in a single sentence using a shared script (e.g., Latin alphabet).
Ambiguous N-grams: Character combinations and short tokens match multiple language profiles simultaneously.
Lack of Single Ground Truth: Standard classifiers expect one primary language label per document rather than multi-label, word-level language switches.

5.What is the difference between language detection and translation?

Feature / Aspect	Language Detection	Machine Translation
Primary Goal	Identifies which language a given text is written in	Converts text from a source language into a target language
Output Type	A language code or label (e.g., 'en', 'hi', 'es')	Converted sentence preserving original semantic meaning
System Role	lassification step at the start of an NLP pipeline	Sequence-to-sequence transformation task
Example Input/Output	Input: "Namaste" -> Output: Hindi (hi)	Input: "Namaste" -> Output: "Hello"

Assignment - 102

1.What are N-grams in NLP?
N-grams are group of continuous words taken from a sentence.

2.What is the difference between unigram, bigram, and trigram?
Unigram - One word group. For example: python
Bigram - Two word group. For example: love python
Trigram - Three word group. For example: I love python

3.Why are N-grams useful in sentiment analysis?
N-grams are useful in sentiment analysis because they capture local word order and context, allowing models to understand multi-word expressions, negations, and phrases (like "not good" or "very happy") that single words alone would miss.

4.How are N-grams different from Bag of Words?

Bag of words	**N-gram**
It focuses on single words	It focuses on word groups
Word order is mostly ignored	Word order is partly maintained
It has less context	It has better context
For Eg: good, not	For eg: not good

5.What can happen if we use very large N-grams on small text data?
If you use very large N-grams on small text data, it leads to severe data sparsity and overfitting, as the model memorizes rare, highly specific word combinations that won't appear in new text.

Assignment - 103

1.What is sentiment analysis in NLP?
Sentiment Analysis is an NLP task used to find emotion or opinion from the text.

2.Why is sentiment analysis useful for businesses?
It helps businesses know whether people are happy, unhappy or neutral about a product or service.

3.What is the difference between positive, negative, and neutral sentiment?
Positive sentiment - Good opinion
Negative Sentiment - Bad opinion
Neutral Sentiment - No strong emotions

4.Why is the word `not` important in sentiment analysis?
The word not changes the meaning. So, the model must understand the full sentence properly.

5.What are some challenges in sentiment analysis?

Challenge	Example
Sarcasm	Great, another delay!
Negation	This movie is not good
Mixed Sentiment	Food was good but service is bad
Emoji	Product is nice 😊
Hinglish Text	Service accha tha but delivery late the

Assignment - 104

1.What is spam detection in NLP?
Spam detection is an NLP task used to identify unwanted, fake promotional, or suspicious messages.

2.Why is spam detection important in email and SMS systems?
We use spam detection to protect users from unwanted messages, fraud links, fake offers, and harmful content.

3.How does Machine Learning help in spam detection?
A Machine learning model predicts whether the message is spam or not.

4.What is the role of TF-IDF in spam detection?
The role of TF-IDF is to convert text into numerical value.

5.What is the difference between false positive and false negative in spam detection?
False Positive - Useful email marked as spam.
False Negative - Spam email marked as normal.

Assignment - 105

1.What is text classification in NLP?
Text Classification is an NLP task where text is divided into different categories.

2.Why do we convert text into numbers before classification?
It helps a computer understand which group a text belong to.

3.What is the difference between binary classification and multi-class classification?
Binary Classification - it has only 2 categories, For example: Spam or Not Spam
Multi-Classification - More than two categories, For example: sports, politics, business

4.Why is spam detection considered a text classification problem?
Because it can detect non spam detection email as spam email.

5.Which algorithms are commonly used for text classification?

Algorithm	Use
Naive Bayes	Good for spam detection and text classification
Logistic Regression	Used for binary and multi-class classification
Support Vector Machine	Good for high-dimensional text data
Random Forest	Used for classification tasks
Deep Learning Models	Used for advanced NLP tasks

Assignment - 106

1.What is a text cleaning pipeline in NLP?
Text cleaning pipeline is a step - by - step process used to clean raw text before using it in a NLP model.

2.Why is text cleaning important before NLP model training?
We use text cleaning pipeline to make text neat, consistent, and useful for NLP model.

3.What are the common steps in a text cleaning pipeline?
Common steps are:

Step	Meaning
Lowercasing	Converts text into small letters
Removing Punctuation	Removes symbols like `!`, `.`, `,`, `?`
Removing Extra Spaces	Cleans unwanted spaces
Removing Stop Words	Removes common words like `is`, `the`, `am`
Tokenization	Breaks text into words
Stemming	Converts words close to root form
Lemmatization	Converts words into proper base form

4.Why should we be careful while removing stop words?
Because removing some words can change the whole meaning of sentence or text.

5.What is the difference between raw text and cleaned text?
Raw text consist of uppercase, punctuation, spaces, stop word, emoji and are in sentences, which is unfit to training NLP model training.
Cleaned text consist of lowercase, no punctuation, no spaces, no stop words, no emoji and are in token format.

Assignment - 107

1.What is TF-IDF in NLP?
TF-IDF is an NLP technique used to convert text into numbers.
TF-IDF gives importance score to words based on how useful they are in a document.

2.What is the difference between TF and IDF?
TF stands for Term Frequency.
IDF stands for Inverse Document Frequency

3.Why does TF-IDF give low value to common words?
TF-IDF assigns a low value to common words because the Inverse Document Frequency (IDF) component mathematically penalizes terms that appear across many documents, signaling that they carry little unique information.

4.What is the difference between Bag of Words and TF-IDF?

Point	Bag of Words	TF-IDF
Meaning	Counts word frequency	Gives importance score
Focus	How many times word appears	How important the word is
Common Words	May give high value	Gives lower value
Output	Word count	Weighted score
Example	`is` may get count	`is` may get low score

5.Why is TF-IDF useful in search engines and resume screening?
Because it is use to finds important keywords and skill.

Assignment - 108

1.What are word embeddings in NLP?
Word embeddings are a way to convert words into numbers.

2.Why do we convert words into vectors?
Because the vectors will help machine to understand the relation between words.

3.What is the difference between one-hot encoding and word embeddings?

Point	One-Hot Encoding	Word Embeddings
Meaning	Represents word using 0 and 1	Represents word using meaningful numbers
Word Meaning	Does not understand meaning	Captures meaning and relationship
Vector Size	Very large	Smaller and dense
Similar Words	Not handled well	Similar words can be closer
Example	`king` and `queen` look unrelated	`king` and `queen` can look related

4.What is Word2Vec?
Word2Vec is a Word Emedding technique used to learn word meaning from nearby words.

5.Why can the word `bank` be difficult for normal word embeddings?
The word `bank` has different meanings.
Old embeddings may give one fixed meaning. Context-based embeddings like BERT can understand the meaning based on the sentence.

Assignment - 109

1.What is cosine similarity in NLP?
Cosine similarity is a method used to how similar two texts, documents, or vectors are.

2.Why do we convert text into vectors before using cosine similarity?
We use cosine similarity to compare text meaning or text matching. It is useful when we want to find whether two sentences, documents, resumes, questions, or search queries are related.

3.What does a cosine similarity score close to 1 mean?
It means the term is very similar.

4.What is the difference between cosine similarity and Euclidean distance?

Point	Cosine Similarity	Euclidean Distance
Meaning	Checks angle between vectors	Checks distance between vectors
Focus	Direction	Actual distance
Used In	Text similarity	Numerical distance problems
Higher Value	More similar	More far
Best For	NLP and document matching	Geometry and distance-based tasks

5.How is cosine similarity useful in resume screening and chatbot projects?
Resume Screening - Matches resume with job description
gChatbot - Matches user question with saved answers.

Assignment - 110

1.What is GloVe in NLP?
GloVe is a word embedding technique used in NLP.

2.What is the full form of GloVe?
GloVe stands for Global Vectors

3.How does GloVe learn word meaning?
GloVe works by checking how often words appear together in a large text collection. This is called word co-occurence. If two words appear near each other many times, GloVe understand that they may bar related.

4.What is the difference between GloVe and Word2Vec?

Point	Word2Vec	GloVe
Meaning	Learns from nearby words	Learns from global word co-occurence
Focus	Local context	Global word statistics
Output	Word Vectors	Word vectors
Use	Word similarity and NLP tasks	Word similarity and NLP task
Example	Learns from word window	Learns from full text pattern

5.Why can words like bank create problems for basic GloVe embeddings?
Because bank has two meaning and need context to refer which meaning it referring so basic GloVe embedding gets confused.

Assignment - 111

1.What is text similarity in NLP?
Text similarity is an NLP technique used to check how similar two pieces of text are.

2.Why do we convert text into vectors before similarity checking?
Because numerical value is easy to compare to each other.

3.What is cosine similarity?
Cosine similarity is one of the most common methods used to measure text similarity.

4.What is the difference between lexical similarity and semantic similarity?
Lexical Similarity - Checks common words. Eg. Python course vs Python training
Semantic Similarity - Checks meaning. Eg, Buy phone vs purchase mobile

5.How is text similarity useful in resume screening and chatbot projects?
In resume screening, it help the model to match the resume with job description.
In Chatbot, it help to find the related knowledge according to the query asked by the user.

Assignment - 112

1.What is text summarization in NLP?
Text summarisation is an NLP technique used to convert long text into a shorter version.

2.Why do we use text summarization?
We use text summarisation to save time and understand long content quickly.

3.What is the difference between extractive and abstractive summarization?
Extractive Summarisation- picks important sentences from original text.
Abstractive Summarisation- Creates a new summary in new words.

4.Why can text summarization be difficult for long documents?
Because the model can miss some important points.

5.How is text summarization useful in news apps and business reports?
Text Summarization useful in news app to create short news summary and business reports uses it for summarizes long report.

Assignment-113

(aug_2nd_2026-topic modeling)

1.What is topic modeling in NLP?
Topic modelling is an NLP technique used to find hidden topics from a collection of text documents.

2.Why do we use topic modeling?
We use topic modelling when we have many documents and want to find the main topics automatically. It is useful to find when reading a

3.Explain the concept of Latent Dirichlet Allocation (LDA) in the context of topic modeling.
LDA assume that a document can contain multiple topics and a topic contain multiple words.

4.What is the difference between topic modeling and text classification?

Point	Topic Modeling	Text Classification
Meaning	Finds hidden topics	Predicts fixed categories
Data Type	Mostly unlabelled data	Labelled data
Output	Topic words	Category name
Example	Finds topic: payment issue	Predicts category: complaint
Learning Type	Unsupervised learning	Supervised learning

5.Why is text cleaning important before topic modeling?
Cleaning is important because if some words appear together many time, the model understand that they may belong to one topic.
For example:
Words: doctor, hospital, medicine, patient
Topic: Healthcare

Words: cricket, match, player, score
Topic: Sports

Assignment - 114

1.What are word embeddings in NLP?
Word embedding are a way to convert words into numbers.

2.Why do we convert words into vectors?
In NLP, computers cannot directly directly understand word like humans. So, words are converted into numerical vectors.

3.What is the difference between one-hot encoding and word embeddings?

Point	One-Hot Encoding	Word Embeddings
Meaning	Represent word using 0 to 1	Represent word using meaningful numbers
Word meaning	Does not understand meaning	Captures meaning and relationship
Vector Size	Very larges	Smaller and dense
Similar Word	Not handled well	Similar words can be closer
Example	King and Queen look unrelated	King and queen can look related

4.What is Word2Vec?
Word2Vec learns word meaning from nearby words.

5.Why can the word `bank` be difficult for normal word embeddings?
The word bank has different meanings. Old embedding may give one fixed meaning. Context - based embedding like BERT can understand the meaning based on the sentence.

Assignment - 115

1.What is Word2Vec in NLP?
Word2Vec is a word embedding technique used in NLP.

2.Why do we convert words into vectors?
In NLP, computers cannot understand words like human, so words are converted to numerical value like vectors.

3.What is the difference between CBOW and Skip-Gram?

Point	CBOW	Skip-Gram
Full Form	Continuous Bag of Words	Skip-Gram
Working	Predicts target word from nearby words	Predicts nearby words from target word
Speed	Faster	Slower than CBOW
Best For	Large datasets	Smaller datasets and rare words
Example	nearby words → target word	target word → nearby words

4.Why does Word2Vec check nearby words?
It follows the idea that words used in similar surroundings usually have similar meanings.

5.What is one limitation of basic Word2Vec?
Its one of the limitation it gets confused when word is polysemy word. For example, the word bank which can refer to river bank and finical bank.

Assignment - 116

1.What is a chatbot?
A chatbot is a computer program that can talk with users through text or voice.

2.Why do we use chatbots in websites and apps?
The chatbot help customers without human support person.

3.What is the difference between rule-based chatbot and AI-based chatbot?
Rule-Based chatbot: Works on fixed rules. For Example: FAQ bot
AI-Based Chatbot: Understands user intent using AI, For example: Smart support bot

4.What is intent in chatbot development?
It defines the purpose of development chatbot, basically define user's purpose.

5.Why is mixed-language text difficult for chatbots?
Mixed language is

Assignment - 117

1.What is a Large Language Model?
A large Language Model is an AI model that understands and generates human like text.

2.Why are Large Language Models important in modern AI?
We use Large Language Models because they can understand natural language and generate useful responses. They help user complete work faster in writing, coding, learning, searching, planning and automation.

3.What is the role of tokens in LLM?
Token are small parts of text used by Large Language Models. A token can be a word, part of word, punctuation mark, or space.

4.What is the difference between NLP and LLM?
NLP helps machine to understand human language
LLM is a model that understands human language and generate text.

5.What are some limitations of Large Language Models?
It can give wrong answers
May not know latest information
Can misunderstand unclear question
Needs careful checking
Can produce biased output

Assignment - 118

1.What is machine translation in NLP?
Machine Translation is an NLP task used to translate text from one language to another language using a computer system.

2.Why is machine translation useful?
It is help in breaking language barrier. It helps people read, write, and communicate in different languages.

3.What is the difference between rule-based translation and neural machine translation?
Rule-Based Machine Translation - Uses grammar rules and dictionaries
Neural Machine Translation. - Uses deep learning models

4.Why is context important in machine translation?
Same word can change meaning in different sentences.

5.What are some challenges in translating mixed-language text?

- **Script & Spelling Mismatch:** Mixed-language text (like Hinglish) uses non-standard phonetic spellings in Latin script rather than official dictionary rules, making standard translation models fail.
- **Lack of Training Data:** Translation engines lack large datasets designed specifically for mixed-language grammar and mixed word pairs.
- **Sentence Structure Confusion:** Models get confused trying to map grammar rules from two different languages mixed inside the same sentence.

Assignment - 119

1.What is a Transformer in NLP?
Transformer is a deep learning architecture used to understand and generate language.

2.Why is attention important in Transformer architecture?
It helps attention to understand this relationship.

3.What is self-attention?
Self-attention is a technique used by transformers to understand the relationship between words in the same sentence.

4.What is the difference between encoder and decoder?
Encoder understand input text.
Decoder generates output text.

5.Why are Transformer models important in modern AI systems?
Transformer models have revolutionized modern artificial intelligence for several major reasons:

- **Parallelization & Scalability:** Unlike older sequential architectures (RNNs), Transformers process entire text sequences simultaneously, making them highly scalable to train on massive GPU clusters over billions of parameters.
- **State-of-the-Art Core Architecture:** Serves as the backbone architecture for almost all leading AI chatbots, code generators, and foundation models (including BERT, GPT series, T5, and modern LLMs).
- **Rich Contextual Understanding:** Excels at recognizing how every word connects to all other words in a document, achieving unprecedented fluency, accuracy, and reasoning in language tasks.

Assignment - 120

1.What is Cell Formatting in Google Sheets?
Cell format means changing the appearance or display style of a cell and its data.

2.What is the difference between Text Colour and Fill Colour?
Text Colour changes the colour of the text inside a cell.
Fill Colour changes the background colour of the cell.

3.What is Text Wrapping and when would you use it?
Text Wrapping controls how long text appears inside a cell. It used wrap allows the complete text to appear inside the cell without making the column extremely wide.

4.What is the difference between a cell's actual value and its number formatting?

Feature / Aspect	Actual Value	Number Formatting
Definition	The underlying raw data stored in memory	The visual presentation shown on the screen
Formula Use	Used by formulas and functions for calculations	Has zero effect on mathematical calculations
Example (Price)	15.8972	Displayed as $15.90 (Currency format)
Example (Date)	45150 (Serial number)	Displayed as 26/08/2026 (Date format)

5.Why should merged cells generally be avoided inside a large data table?
It should generally be avoided inside normal data tables because it can create problems while sorting and filtering data.

Assignment - 121

1.How can you create a new Google Sheet?
Creating a new sheet means creating a new spreadsheet where we can enter, organise and manage data.

2.What is the difference between a spreadsheet and a sheet?
A spreadsheet is the entire file or computer program (like Microsoft Excel or Google Sheets) used to organise and analyse data. A sheet (or worksheet) is an individual page or tab inside that spreadsheet file.

3.How can you add and rename a new sheet tab?
A new sheet can be added inside an existing spreadsheet, at the bottom of Google sheets, Click the + button, a new sheet will be created.
Sheet tabs should be renamed according to the data they contain, steps :

- Go to the sheet Tab.
- Double-click the sheet name or open its menu.
- Select Rename.
- Enter the new name.
- Press Enter.

4.Why is the Duplicate Sheet option useful?
It is useful when we want the same structure or format for another month department or project.

5.How would you organise a new sheet for maintaining monthly employee attendance?
Create a table with employee names in the rows and dates (1 to 31) across the columns to mark daily status like Present, Absent, or Leave.

Assignment - 122

1.What is Date and Time Formatting in Google Sheets?
Date and Time formatting means changing how a date or time is displayed inside a cell.

2.What is the difference between `TODAY()` and `NOW()`?
The TODAY() function returns the current date.
The NOW() function returns the current data and time.

3.What is the difference between 12-hour and 24-hour time formats?
12-Hour format, it uses AM and PM. 24-Hour format, it does not require AM or PM.

4.How can you calculate the number of days between two dates in Google Sheets?
Google Sheets can calculate the number of days between two dates.
Suppose:
A2 = 18/08/2026
B2 = 25/08/2026
Formula:
=B2-A2
Result:
7

5.If a company wants to calculate employee working hours using Check-In and Check-Out time, how can it be done?
We can calculate working duration using:
=Check-Out - Check-In
For actual cell references:
=D2-C2

Assignment - 123

1.How do you enter and edit data in Google Sheets?
To enter data:

- Select a cell
- Type the required value
- Press Enter

2.What is the difference between Copy and Cut?
Copy: Creates a copy of selected data.
Ctrl + C
Cut: Moves selected data.
Ctrl + X

3.What is the Fill Handle and where can it be useful?
The fill Handle is used to quickly copy data or continue a pattern. It appears at the corner of a selected cell or range.

4.How can Find and Replace save time while working with large data?
Instead of editing every cell manually, find and Replace can update the matching values.

5.If you accidentally replace important data in a cell, what can you do to recover the previous value?
We can use undo by pressing ctrl + Z.

Assignment - 124

1.What are the main parts of the Google Sheets interface?
The main part of the interface are:
Menu Bar
Toolbar
Formula Bar
Name Box
Rows
Columns
Cells
Sheet Tabs
Status Area

2.What is the difference between the Name Box and Formula Bar?
The name Box shows the address of the currently selected cell.
The formula Bar is used to view, enter and edit the content or formula of the selected cell.

3.What is an Active Cell in Google Sheets?
The Active Cell is the cell that is currently selected.
When you click a cell, a border appears around it.
Any data you type willl base entered into the active cell.

4.What are Sheet Tabs and why are they useful?
Sheet Tabs are available at the bottom of the spreadsheet. They allow us to create multiple sheet inside one Google sheets file.
A company can maintain:
Sheet 1 → Employees
Sheet 2 → Attendance
Sheet 3 → Salary
Sheet 4 → Expenses
All these sheets can exist inside the same spreadsheet file.

5.If a spreadsheet has thousands of rows, how would you keep the heading row visible while scrolling?
We can use Freeze, keep selected rows or columns visible while scrolling.

Assignment - 125

1.What is Google Sheets?
Google Sheet is an online spreadsheet application provided by Google.

2.What is the difference between a row, column and cell?
Row are horizontal and represented by number.
Columns are vertical and represent by letters.
Cell is the box created where a row and column meet.

3.What is a cell range? Give one example.
A range is a group of cells selected together.
For example:
A1:A10

4.What is the difference between a formula and a function in Google Sheets?
A formula is used to perform calculations automatically. A formula normally starts with the = sign.
For Example:
=A1+B1

Functions are ready-made formulas available in google sheets.
For Example:
SUM(A1+B1)

5.If multiple employees need to update the same spreadsheet at the same time, how can Google Sheets help?
Real-Time Co-Authoring: Multiple users can view, edit, and enter data simultaneously inside the exact same document without file lock issues.
Live Cursor Tracking: Displays color-coded user cursors and highlights active cells showing who is editing in real time.
Integrated Communication: Supports cell-level commenting, `@mentions`, and action items to assign tasks and discuss edits directly inside the file.
Version History & Control: Automatically tracks every change with full user attribution, allowing teams to view or restore prior document versions anytime.

Assignment - 126

1.What is Number Formatting in Google Sheets?
Number Formatting means changing how a number is displayed in Google Sheets.

2.What is the difference between Currency and Percentage Formatting?
Currency Formatting displays a number as money.
Percentage Formatting displays a decimal value as a percentage.

3.What is the difference between the actual value of a cell and its displayed value?
Actual value: The real data used in calculation. For eg: 12.390 or SUM(A1:B1)
Display Value: The Formatted visual output shown in the grid. For eg:55%, $359.

4.If a cell contains `0.75`, what will it display after applying Percentage Formatting?
It will display 75%.

5.Why can a number stored as text create problems while performing calculations?

- **Formula Ignorance:** Mathematical aggregation functions like `SUM`, `AVERAGE`, and `COUNT` ignore text-formatted numbers completely, returning incorrect totals.
- **Calculation Errors:** Direct arithmetic operations may return `#VALUE!` errors or unexpected concatenated results instead of mathematical sums.
- **Sorting & Lookup Failures:** Text numbers sort alphabetically (e.g., `"10"` comes before `"2"`) rather than numerically, breaking data sorting and precise `VLOOKUP` or `XLOOKUP` matches

Assignment - 127

1.What is the difference between a row, column, and cell in Google Sheets?
Row runs horizontally.
Column runs vertically.
Cell box created where a row and column meet

2.What is a cell address? Give two examples.
Every cell has a unique address based on its column and row.

3.What is a cell range and how is it represented?
A Cell range is a group of cells selected together. A colon is used to represent a range.
Example:
A1:A10

4.What is the difference between deleting the content of a cell and deleting an entire row?
Deleting the content of a cell refers to the removing the data of a particular cell. And deleting an entire row refers to the removing the entire horizontal line along the data it consist.

5.If you have employee data, how would you organise it using rows and columns?
I would organise the data using.

	Name of employee	City	Salary	Job role	Joined Date
1	Aman	Delhi	30000	Frontend Developer	01.09.2023
2	Riya	Noida	35000	Backend Develper	23.09.2022

Assignment - 128

1.What is Text Formatting in Google Sheets?
Text formatting means changing the appearance of text inside a cell. It helps us make important information clear and keeps the sheet organised.

2.What is the difference between Bold, Italic and Underline?
Bold makes the text stronger and darker.
Italic makes the text slightly slanted.
Underline adds a line below the text.

3.What is Text Rotation and where can it be useful?
Text Rotation changes the direction or angle of text inside a cell.
It can be useful when:

- Column headings are long
- Many columns are present
- We want to save horizontal space

4.What is the difference between Text Formatting and Cell Formatting?

Text Formatting	Cell Formatting
Mainly changes text appearance	Can change the complete cell appearance
Font Style	Fill Colour
Font Size	Borders
Bold	Number Format
Italic	Cell background
Underline	Merge Cells
Text Colour	Other cell-level formatting

5.How would you format the heading row of an employee report to make it easy to read?
1.Apply Bold Styling:
Select the entire header row and press **Ctrl + B** to make the column titles distinct from the data rows.
2.Set a Distinct Fill Color:
Apply a dark background fill color (e.g., dark blue or slate grey) to the header cells to separate them visually from the grid.
3.Adjust Text Color for High Contrast:
Change the text color to white or light yellow so it contrasts sharply against the dark background.
4.Center Align & Wrap Text:
Center-align the header labels both horizontally and vertically, and enable **Text Wrapping** so long titles display cleanly.
5.Freeze the Header Row:
Go to **View -> Freeze -> 1 row** to keep the formatted headers visible at all times while scrolling down.

Assignment - 129

1.What are errors in Google Sheets?
An error appears when Google Sheet cannot properly calculate or understand a formula.

2.What is the difference between `#N/A` and `#REF!`?
#N/A means the required value is not available or not found.
#REF! Means the formula contains an invalid cell reference.

3.When does `#DIV/0!` appear?
This error appears when we try to divide a number by zero.

4.Why does `#VALUE!` appear?
It appears when the formula receives the wrong types f data.

5.What is the use of `IFERROR` in Google Sheets?
It is use to handle the error. It is use to check error.

Assignment - 130

1.What is HLOOKUP?
HLOOKUP is used to search for data horizontally in Google Sheets.

2.What does H mean in HLOOKUP?
H in HLOOKUP stands for Horizontal.

3.What is the basic formula of HLOOKUP?
=HLOOKUP(search_value, range, row_number, FALSE)

4.What is the difference between VLOOKUP and HLOOKUP?

VLOOKUP	HLOOKUP
Searches vertically	Searches horizontally
Searches in first column	Searches in first row
V = Vertical	H = Horizontal

5.What does `FALSE` mean in an HLOOKUP formula?
FALSE means find the exact value.

Assignment - 131

1.What is a Pivot Table in Google Sheets?
A pivot Table is used to sumarise and analyse large data in Google sheets.

2.Why do we use Pivot Tables?
Pivot Tables helps us understand large data quickly.

3.What are Rows and Values in a Pivot Table?
Rows decide how we want to group the data.
Values are the number we want to calculate

4.What is the use of Filters in a Pivot Table?
Filter are used to show only the data we need.

5.How would you use a Pivot Table to calculate total sales by city?
We can use SUM to calculate the total sales.

Assignment - 132

1.What is VLOOKUP?
VLOOKUP is used to search for data verticals in Google Sheets.

2.What does V mean in VLOOKUP?
V in VLOOKUP stands for Vertical.

3.What is the basic formula of VLOOKUP?
=VLOOKUP(search_value, range, column_number, FALSE)

4.What is the difference between VLOOKUP and HLOOKUP?

VLOOKUP	**HLOOKUP**
Searches vertically	Searches horizontally
Searches in first column	Searches in first row
Returns data from the right	Returns data from a row below
V = Vertical	H = Horizontal

5.What does `FALSE` mean in a VLOOKUP formula?
FALSE means find the exact match.

Assignment - 133

1.What is the IF Function in Google Sheets?
The IF function is used to check a condition and return a result based on whether the condition is TRUE or FALSE.

2.What is the syntax of the IF Function?
=IF(condition, value_if_true, value_if_false)

3.What is the difference between > and >=?
> mean to Greater than and >= Greater than or equal to.

4.How can you use IF to show Pass or Fail based on marks?
=IF(B2>=40,"Pass","Fail")

5.Can the IF Function check text values? Give an example.
=IF(A2="Active","Working","Not Working")

Assignment - 134

1.What is the SUM Function?
The SUM Function is used to add numbers in Google Sheets.

2.What is the SUMIF Function?
It is used to add numbers based on one condition.

3.What is the difference between SUM and SUMIF?

SUM	SUMIF
Adds all selected numbers	Adds numbers based on a condition
No condition required	One condition is required
SUM(B2:B4)	SUMIF(A2:A4,"Delhi",B2:B4)

4.Explain the syntax of SUMIF.
=SUMIF(range, criteria, sum_range)
Easy meaning:
=SUMIF(Where to Check, What to Check, What to Add)

5.How would you calculate total sales only for Delhi using SUMIF?
Suppose we have:

City	Sales
Delhi	5000
Patna	7000
Delhi	3000
Patna	2000

We want to calculate only **Delhi Sales**.
Formula:
=SUMIF(A2:A5,"Delhi",B2:B5)
Result:
8000
Because:
Delhi → 5000
Delhi → 3000
Total → 8000

Assignment -135

1.What are input fields in Streamlit?
Input fields allow users to enter information into an application.

2.What is the purpose of `st.form()`?
st.form() is like a form where we have to fill multiple details. It is group of multiple input.
It submits the data only after the user clicks the button.

3.What is input validation?
Input validation checks whether the entered data is correct and acceptable.

4.What is the difference between `st.selectbox()` and `st.number_input()`?
st.selectbox() selects one options.
st.number_input() accepts numbers.

5.How can you display validation errors in Streamlit?
If errors are available, display them using st.error().

Assignment - 136

1.What is basic prediction with an IF condition?
Basic prediction uses python if, elif, and else conditions to calculate an estimated result.

2.What are the three inputs used in this application?
In this application will collect only three details from the user which predicts the price of the used car. The 3 inputs are as follows:
Car Brand - set the starting price
Manufacturing Year- calculates the car age
Kilometres Driven - Reduces the price based on usage

3.What is the difference between `if`, `elif` and `else`?
An if conditions runs code when a given condition is true.
Elif checks another condition when the previous condition is false.
else runs when none of the previous conditions are true.

4.How is the predicted car price calculated?
The predicted price calculated using the following formula:
Predicted_Price = Base Price - Age Depreciation - KM Depreciation

5.What is the difference between basic IF prediction and Machine Learning prediction?

Basic IF Prediction	**Machine Learning Prediction**
Uses manually created rules	Learns from a dataset
Uses if, elif and else	Uses a trained model
Does not require training	Requires model training
Developer decides the price rules	Model learns price patterns

Assignment - 137

1.What are missing values in a dataset?
A missing value means information is not available in a dataset cell.

2.How do you check missing values using Pandas?
We can check missing value using the following command:
df.isnull().sum()

df -> data frame
isnull()-> checks every cell for missing value.
sum() -> counts missing values in each column.

3.What are duplicate records?
Duplicate records are repeated rows in a dataset.

4.How do you count duplicate rows in a DataFrame?
We can count duplicate rows in a DataFrame by using following command:
df.duplicated().sum()

Df-> data frame
Duplicated-> it is function used to find duplicate
sum -> it is the function used to count the duplicates

5.Why should we check missing values and duplicates before model training?
It is important to check missing values and duplicates before training the model because missing values and duplicate values can affect predictions of model negatively.

Assignment -138

1.What is Streamlit?
Streamlit is the tool to build professional looking website.

2.Why do we create a virtual environment?
We create virtual environments so that different projects can run different versions they required and create the requirement.txt file during deployment this file is needed.

3.Which command is used to install Streamlit?
pip install streamlit

4.How do you run a Streamlit application?
streamlit run file_name.py

5.What is the purpose of the `app.py` file?
app.py is streamlit application.

Assignment - 139

1.What is the purpose of `login.py`?
The login.py will add a login page before accessing the page.

2.What is st.session_state?
st.session_state is used to stores the login status.

3.How do you hide a password in Streamlit?
We can hide the password using st.text_input(”Password”, type = ”password”)

4.What is the purpose of `st.stop()`?
It is used to refreshes the application after login or logout.

5.Why should passwords not be stored directly in code for a real application?
So that password can only known to user.

Assignment - 140

1.What is a rule-based chatbot?
The rule-based chatbot does not use artificial intelligence or an external API. It uses keyword conditions. It reads information directly from datasets. If the dataset changes, the chat calculations also change automatically.

2.What is the purpose of `chatbot.py`?
The chatbot.py file is used to add chatbot to the application.

3.How does the chatbot read information from the dataset?
It uses the pandas library to read the dataset.

4.What is the purpose of `st.chat_input()`?
It purpose is to get user's question.

5.How does `st.session_state` maintain chat history?
1- Initialize List Key
2-Append User Prompts & Bot Response
3- Re-render Message History

Assignment - 141

1.What is categorical data?
Categories data contains names, labels or groups instead of continuous numerical values.

2.Why is categorical data encoding required?
Machine Learning models normally work wth numerical values.
Linear Regression cannot directly understand text such as:
Maruti
Petrol
Mannual
Therefore, we convert text categories into numerical columns.
This process is called Categorical Data Encoding.

3.What is One-Hot Encoding?
One Hot Encoding creates a seperate numerical column for category.

4.What is the purpose of `ColumnTransformer`?
It is used to encode categorical columns and keep numerical columns unchanged.

5.What does handle_unknown="ignore" mean?
When defining `OneHotEncoder(handle_unknown="ignore")`, it instructs the encoder to handle unseen categorical labels encountered during testing or real-time inference gracefully.

Why it is essential:

- **Standard Behavior (`error`):** If a model is trained on brands `[Maruti, Hyundai]` and receives a new testing input with brand `[Honda]`, the default encoder raises an execution error and crashes.
- **With `handle_unknown="ignore"`:** Instead of throwing an error, the encoder sets all one-hot encoded binary columns for that unknown category to `0` (e.g., `Fuel_Is_Petrol=0`, `Fuel_Is_Diesel=0`), allowing the model to make predictions safely without system failure.

Assignment - 142

1.What is Exploratory Data Analysis?
Exploratory Data Analysis is the process of studying and understanding a dataset.

2.Why is EDA performed before model training?
Because it help us select columns columns for machine learning.

3.What information does `df.describe()` provide?
Use df.describe() to view the statistical summary of numerical columns.

4.How do you calculate the average selling price by brand?
df[‘selling_price’].mean()

5.What is correlation in Data Science?
Correlation show the relationship between numerical columns.

Assignment - 143

1.What is a User Interface?
User Interface means the visible part of an application.
It is also called UI. It is the frontend of the website.

2.Why is CSS used in a Streamlit application?
CSS is used in a Streamlit application to customize the default styling, layout, and visual elements beyond built-in themes. It allows developers to apply custom branding, adjust spacing, modify colors, and create unique UI designs for their web apps.

3.What is the purpose of `st.columns()`?
Helps to divide the pages. It helps divide the web page in different sections providing structure.

4.What does unsafe_allow_html=True do?
Allows Streamlit to display HTML and CSS.

5.Why should UI code be kept in a separate file?
So that the design remain same for separate we page.

Assignment - 144

1.What are features in Machine Learning?
Features are the input columns used by a Machine Learning Model

2.What is a target column?
The target is the value that machine model has to predict.

3.What is the difference between X and y?
Features are normally stored in X.
Target is normally stored in y.

4.Why should the target column not be included in X?
The capital letter X represents multiple inputs columns.

5.What is the difference between categorical and numerical features?
In Categorical features, columns contains text value.
In Numerical features, columns contains numerical values.

Assignment -145

1.What is Linear Regression?
We will train a Linear Regression model to predict the selling price of a used car.

2.What is the purpose of `train_model.py`?
It import the dataset, select the feature, selects the target, divide data into training data and testing parts, and trains the ML model and the model as .pkl file.

3.What is the purpose of `app.py`?
It loads the saved models, collect three car details, create an input table, send the input to the model, and displays the predicted values.

4.What is stored inside a `.pkl` file?
It stores the trained model.

5.What is the difference between `fit()` and `predict()`?
fit() is used to trains models and predict() is used to predicts a target value.

Assignment - 146

1.What is the `datetime` module in Python?
Datetime is a built-in python mode;e used for working with date and time.

2.What is the difference between `date`, `time` and `datetime`?
date - works with date.
time - work with time.
datetime- works with both date and time.
timedelta - works with date/time difference.

3.What is the difference between `strftime()` and `strptime()`?
strftime() - format date and time
strptime() - Convert string into datetime

4.What is `timedelta` used for in Python?
It used to add or subtract time.

5.How can you get the current date and time in Python?
We can use datetime.now() to get the current data and time in python.

Assignment - 147

1.What is a module in Python?
A module is a python file that contains reusable code.

2.Why are modules used in Python?
Modules help us to reuse code, keep code organized, reduce repeated code, and make programs easier to manage.

3.How do you import a module in Python?
We use the import keyword.
Syntax:
```
import module_name
```
For example:
```
import math
```

4.What is the difference between `import module` and `from module import function`?
We use the import keyword, which loads the whole library.
from module import function is way load only a particular function.

5.What is an alias in Python modules?
We can give a shorter name to a module using as.
For example:
```
import math as m
print(m.sqrt(49))
```

Assignment - 148

1.What is a package in Python?
A package is a folder that contains multiple python modules. It is used to organize related python files in one place.

2.What is the difference between a module and a package?

Module	Package
A Python file	A folder
Contains Python code	Contains modules
Example: `add.py`	Example: calculator/
Used for specific functionality	Used to organize multiple modules

3.What is the purpose of `__init__.py`?
It is a special python file commonly used inside a package.

4.How do you import a function from a module inside a package?
Instead of importing only a function, we can import the complete module.

5.How do you install an external Python package using pip?
External packages can be installed using pip.
Syntax:
pip install package_name

Example:
pip install pandas

Another example:
pip install numpy

Assignment -149

1.What is a README file?
A README file explain the project.

2.What is Markdown?
Markdown is used to format text using symbols.

3.How do you create headings in Markdown?
The hash '#' is used to create headings.
For example:
Main Heading
O/p:
Main Heading

4.How do you add a link to a README file?
The syntax of adding a link is as follows:
[Link Name] (Link Address)
For Example:
[Streamlit] (https://streamlit.io)

5.What information should be included in a project README?
It should include the following:

- Main heading (usually the name of the project)
- The details of the project.
- Features of the project
- Input fields
- Technologies used
- Project Structure
- File Information
- Step to install the project
- How to run the project
- Workflow
- Important note
- Useful Links
- Creator's details
- Type of project

Assignment - 150

1.Why are charts used in a Data Science application?
It helps in visualizes the data, so we can understand the data better and analy it sufficient

2.What is the difference between a bar chart and a line chart?
Bar chart is to compare different values.
Line chart is to display changes over time.

3.How do you create a chart in Streamlit?
The syntax of creating data:
st.bar_chart(data_set)

4.How do you calculate the average selling price by brand?

```
brand_price = (
    df.groupby("brand")["selling_price"]
    .mean()
    .sort_values(ascending=False)
)
```

5.What is the purpose of `st.tabs()`?
It is used to create chart tabs.

Assignment - 151

1.What is a TOML file?
TOML is a file format used to store applications settings.

2.What is the purpose of `config.toml` in Streamlit?
config.toml is the main configuration file used by streamlit. It allows us to change the application theme without writing css.

3.Where should the Streamlit configuration file be stored?
Inside a folder called .streamlit.

4.How does Streamlit read `config.toml`?
Without config.toml, Streamlit uses its default theme.
With config.toml, we can create a theme for the website.

5.What is the difference between `config.toml` and `app.py`?
TOML is a config file, it use for theme settings, visual parameter and server setting. It is automatically realized by streamlit so it doesn't need the
app.py is a python script. It is used for core application logic, ML model loading, user input, and page routing.

Assignment - 152

1.Why do we display an expected price range?
So Machine Learning model give better result to the user.

2.How is the minimum expected price calculated?
Minimum price = Predicted Price - 10%

3.How is the maximum expected price calculated?
Maximum Price = Predicted Price + 10%

4.What is the purpose of `st.metric()`?
st.metric() display price information.

5.What factors can change the actual selling price of a car?
Car condition
Service history
Accident history
Insurance
City
Market demand
Tyre condition

Assignment - 153

1.What is SQLite?
SQLite is a small database system included with python.

2.What is the purpose of `database.py`?
It creates and manages the database.

3.How does the login system check a user?

```
def authenticate_user(username, password):
    """
    Check username and password.
    """
    connection = sqlite3.connect(
        DATABASE_NAME
    )
    cursor = connection.cursor()
    password_hash = hash_password(
        password
    )
    cursor.execute(
        """
        SELECT id, username
        FROM users
        WHERE username = ?
        AND password_hash = ?
        """,
        (
            username,
            password_hash
        )
    )
    user = cursor.fetchone()
    connection.close()
    return user
```

4.Why should passwords be stored as hashes?
To keep the password safe and secure.

5.What is a parameterised SQL query?
The `?` symbols create a parameterised query:

```
WHERE username = ?
AND password_hash = ?
```

This is safer than joining user input directly with SQL.

Assignment - 154

1.What is model evaluation?
Model evaluation checks how well the Linear Regression model predicts car prices.

2.What is the difference between MAE and MSE?
MAE -> Mean Absolute Error. MAE calculates the average difference between actual and predicted prices.
MAE = Sum of | Actual Price - Predicted price| ÷ Total Predictions
MSE -> Mean Square Error . MSE calculates the average squared difference.
MSE = Sum of (Actual Price - Predicted Price)² ÷ Total Predictions

3.What does RMSE represent?
RMSE is the square root of MSE
RMSE is easier to understand because it uses the same unit as the selling price.
Formula:
RMSE = Square root of MSE

4.What does the R^2 score explain?
R^2 shows how well the model understands price changes.
The score is normally checked between :
0 to 1
For example:
R^2 = 0.85

5.Why do we compare actual and predicted prices?
We compare actual and predicted prices to get prediction value.

Assignment - 155

1.What is a `.gitignore` file?
A .gitignore file tells Git which files and folder should not be uploaded to GitHub.

2.Why should the virtual environment not be uploaded to GitHub?
Virtual environment consist of large binary dependencies that are system-dependent and are easily recreated on any computer using a requirement.txt file, making them redundant and wasteful to upload.

3.How do you ignore all `.pyc` files?
To ignore all complied python bytecode (.pyc) files across all the project subdirectories, add *.pyc to your .gitignore file.

4.Why should `.env` and `secrets.toml` be ignored?
The file .env and secret.toml contain confidential credentials, Private API keys, and database passwords that must be kept secret to prevent unauthorized access or security leaks.

5.What does git rm -r --cached venv do?
It removes the venv directory from git tracking cacfewhile keeping the local files intact on computers.

Assignment - 156

1.What is DAX, and where do you enter a DAX formula in Power BI?
Dax stands for Data Analysis Expressions. It is a formula language used to perform calculations on data loaded into Power BI.

2.What is the difference between a calculated column and a measure?

Calculated Column	Measure
Calculated for every row	Calculates one result
Stored in the data model	Calculated when required
Can be used in a slicer	Mainly used in the Values section
Does not change for each row filter	Changes according to report filters
Created using New column	Created using New measure

3.What is the purpose of the `CALCULATE` function?
A calculated column create a new column in the table. Power Bi calculates its value separately for every row.

4.Why does the Total Sales measure change when a city is selected in the slicer?
The Total_Sales measure uses a simple aggregation formula: SUM(Sales[Sales_Amount]).
Measures respond dynamically to the report's Filter Context. When a user selects a specific city (such as Mumbai or Delhi) in a slicer,PowerBI filters the underlying table down to only those matching row. The measure then recalculates the sum strictly for the filtered dataset.

5.Why can the Delhi Sales measure continue to show Delhi's value when another city is selected?
The Delhi Sales measures is written using the calculation function with an explicit hardcoded filter:
Delhi_Sales = CALCULATE([Total_Sales], Sales[City] = "Delhi")
The CALCULATE function overrides and replaces any conflicting filter context coming from report slicers for the CITY column. Therefore, even if a user selects another city (like " Mumbai") in a slicer, CALCULATE enforce Sales[City] = "Delhi", ensuring it evaluates only Delhi's sales data.

Assignment - 157

1.What is a CSV file, and how is it imported into Power BI?
CSV stands for Comman Separated Values. It is a text file used to store data in row and columns.
To import a CSV file into Power Bi Desktop.

- Open Power BI Desktop
- Select the Home tab from the top ribbon
- Select Get data.
- Select Text/CSV

If Text/CSV is not directly visible:

- Select Get Data -> More
- Open the File category
- Select Text/CSV
- Select Connect

2.What is the difference between Load and Transform Data?
We select Load is when column name are correct, Data types are correct, there are no unwanted rows, and, the files does not require cleaning.
We select Transform Data when column names need to be changed, Data types are incorrect, blank rows are present, Some columns are not required, and, The data contains error.

3.How can you change the data type of a CSV column in Power Query?
Here are the following steps:

- Select the columns
- Select the data type icon beside the column name
- Choose the required data type
- Select Replace current if a message appears

4.What happens during refresh if the original CSV file has been moved?
It will show a location error.

5.How can changing a CSV column name affect an existing Power BI report?
We can use Transform Data, it step are as follows:

- Select Transform Data in the preview window
- Power Query Editor will open
- Find the query in the Queries pane on the left
- Right-click the query name
- Select Rename
- Enter name you want your column to be
- Then press Enter

Assignment - 158

1.What is the role of a pandas DataFrame in Power BI?
Power BI import Python data only when it is available as a pandas DataFrame.

2.How can you import a CSV file into Power BI using Python?
Python can import data from files such as CSV and Excel. The imported data must be stored in a pandas DataFrame.
Here are steps are as follows:

- Open Power BI Desktop
- Select Home -> Get Data
- Select Other -> Python script
- Enter the Python code
- Click OK
- Select the required DataFrame in the Navigator window
- Select Load or Transform Data

3.What is the difference between using Python under Get Data and using it in Power Query?
Python can import data from files such as CSV and Excel. The imported data must be stored in a pandas DataFrame.
Get Data uses the Text/CSV connector to import a CSV file into Power BI to import a CSV file into Power BI Desktop.

4.Why might a successfully created Python variable not appear in the Power BI Navigator?
Because it should be in DataFrame otherwise it will not appear in the Power BI Navigator.

5.What can happen during refresh if the Python environment used by Power BI does not contain pandas?
The Data would not be loaded to Power BI

Assignment - 159

1.What is Power BI?
Power BI is a data analysis and visualization tool developed by the microsoft.
It is used to create:
Reports
Charts
Tables
Dashboards

2.What is a dataset?
A dataset is a collection of related information arranged in rows and columns.

3.What is a CSV file?
It is stands for Comma Separated Value. It is a file format used to store data in rows and columns.

4.How can you import CSV data into Power BI?
Go to Home of the Power Bi
Click on Get data option on the top bar
Then Select Text/CSV

5.What is the difference between a report and a dashboard in Power BI?
Report means data is represented in a structure manner rows and columns.
Dashboard means data is visualized using charts, pie charts, etc.

Assignment - 160

1.What is a dataset?
A dataset is a collection of related information arranged in rows and columns.

2.What is the difference between a row and a column?
A row is also called a record.
A column is also called a field

3.Why is Company_ID required in this dataset?
Every Company has unique Id when companies can have same name then they can be identified by their company's Id

4.How can you import a CSV dataset into Power BI?
Go to Home of the Power Bi
Click on Get data option on the top bar
Then Select Text/CSV

5.Which data types should be used for Employees and Annual_Revenue_Cr?
Employees is Whole number.
Annual_Revenue_Cr is whole number. According to the this project.

Assignment - 161

1.How do you import a CSV dataset into Power BI?
Go to Home of the Power Bi
Click on Get data option on the top bar
Then Select Text/CSV

2.What is the difference between Load and Transform Data?
Load means importing the dataset directly. It is used when dataset is clean and doesn't consist of any error.
Transform Data open the dataset in Power Query. It is used when dataset is not clean and consist of missing value, duplicate values, etc.

3.Why should data types be checked before creating a dashboard?
Because if data types are not correct then visualization of the would be incorrect and inaccurate.

4.What is the purpose of Close & Apply?
It will close the power query and apply the changes to the data. After that data will appear in the Data pane.

5.How can duplicate company records be removed in Power Query?
Open the power query editor
Click on the header of the company that contains the company name or company Id.
Go to the Home tab on the ribbon.
Click Remove Rows and Select Remove Duplicates.
Power Query keeps the first row it finds for each unique company and deletes the rest.

Assignment - 162

1.What is a pie chart in Power BI?
A pie chart displays data as slices of a circle.

2.When should you use a pie chart?
A pie chart is useful when:

- We want to show parts of a total
- The dataset has a small number of categories
- We want to compare percentages
- The category names are clear

3.Which fields are used to compare IT and Non-IT companies?
Legend -> Company_Type
Values -> Sum of Annual_Revenue_Cr

4.How can you display percentages on a pie chart?
Click on pie chart to select it
Go to the visualization pane and click the format icon.
Scroll down and find the detail labels option, then turn it on.
Expand the Detail labels menu.
Look for label style or contents.
Choose Percentage, Category and percentage, or All details from the dropdown menu.

5.What is the difference between a pie chart and a donut chart?

Pie Chart	**Donut Chart**
Complete circle	Circle with an empty centre
Displays slices	Displays slices in a ring
Shows parts of a total	Shows parts of a total
Suitable for fewer categories	Can display a centre value

Assignment - 163

1.What is a report page in Power BI?
A page is a separate working area inside a power BI report.

2.Why are different pages used in a Power BI report?
Different pages helps us organize the report.

3.What is a line chart?
A line chart displays data points connected by a line.

4.Which fields are used to create the company trend line chart?
X axis and Y axis line is used to find the trend line .

5.How can you display separate lines for IT and Non-IT companies?
To display separate lines for IT and Non-IT companies:
X-axis → Founded_Year
Y-axis → Count of Company_ID
Legend → Company_Type
The chart will display:
Blue Line → IT Companies
Orange Line → Non-IT Companies

Assignment - 164

1.What is a card visual in Power BI?
A card visual displays an important value from the dataset.

2.What is the difference between Count and Distinct Count?
Count includes all the value.
Distinct Count ignores the duplicate values.

3.How can you create a Total Employees card?
Click on new measured.
Create total employees :
Total Employees =
SUM(Companies[Employees])
Then add a cared visual
Drag the measure into values.

4.Why are DAX measures useful for card visuals?
DAX measures provide clear names and reusable calculations.

5.How do slicers affect card values?
Card values can change when a slicer is used.